

Résultats d'analyses

Analyses de diversité alpha

Plot et significativité de q=0 et q=1

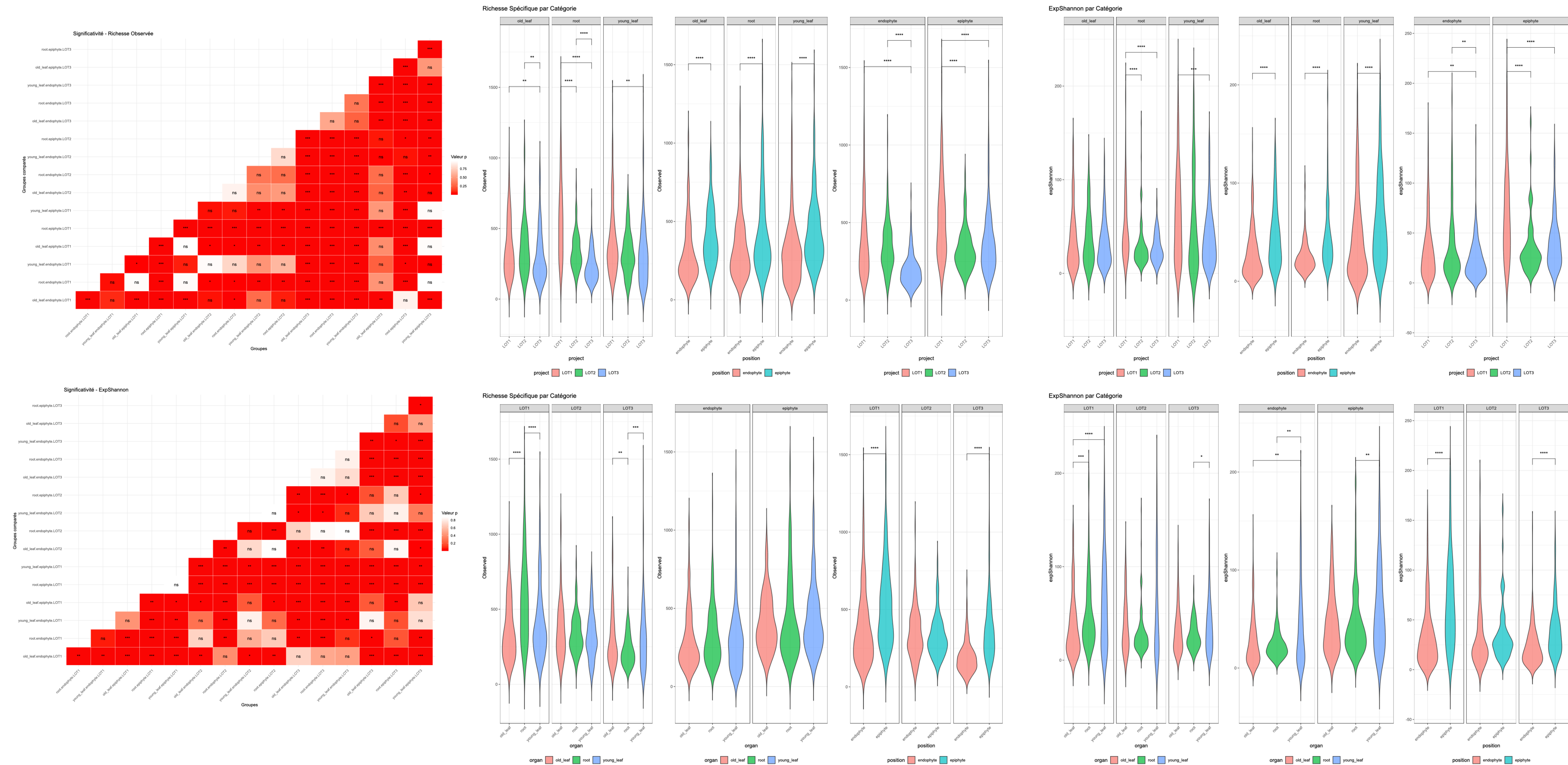
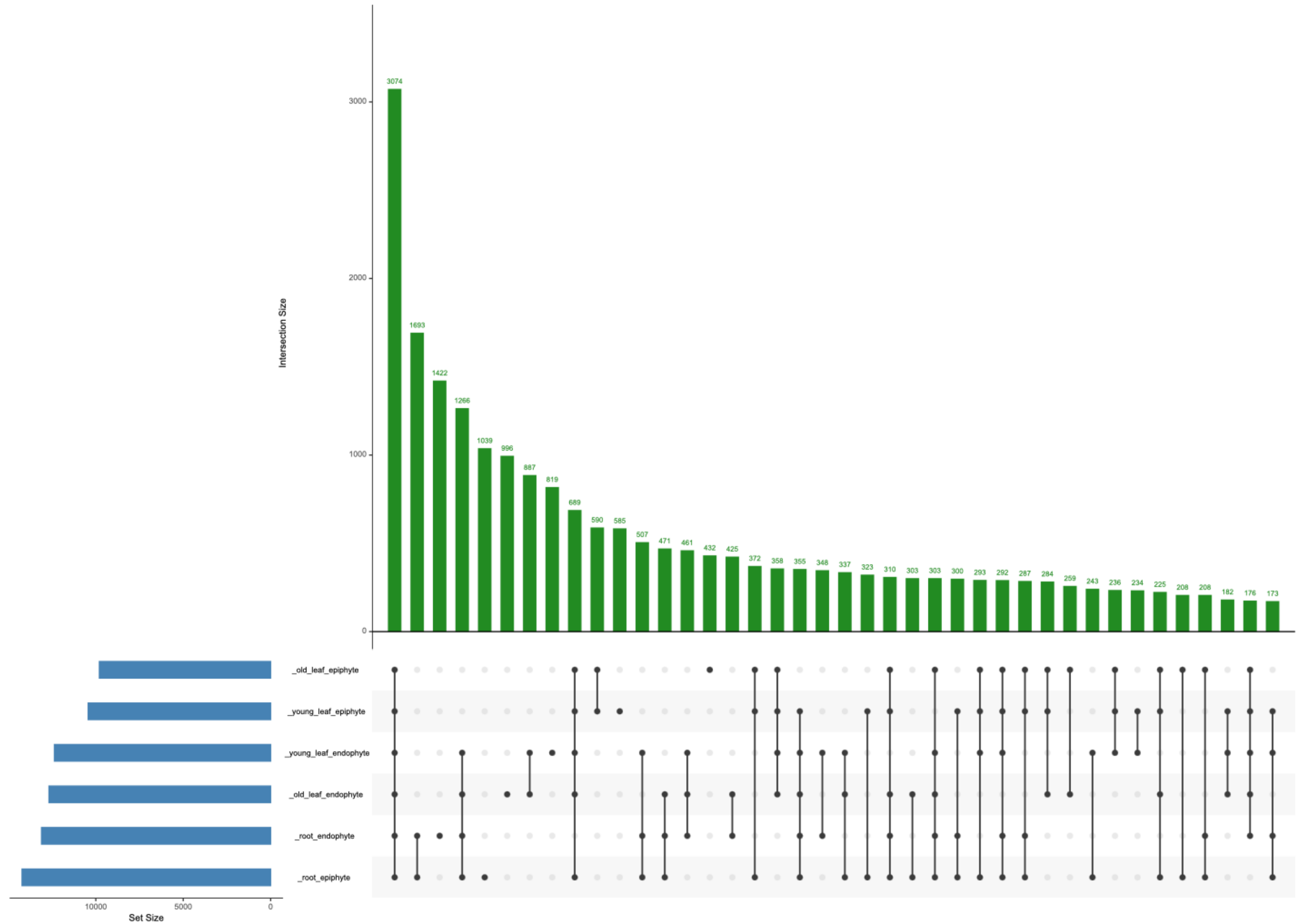
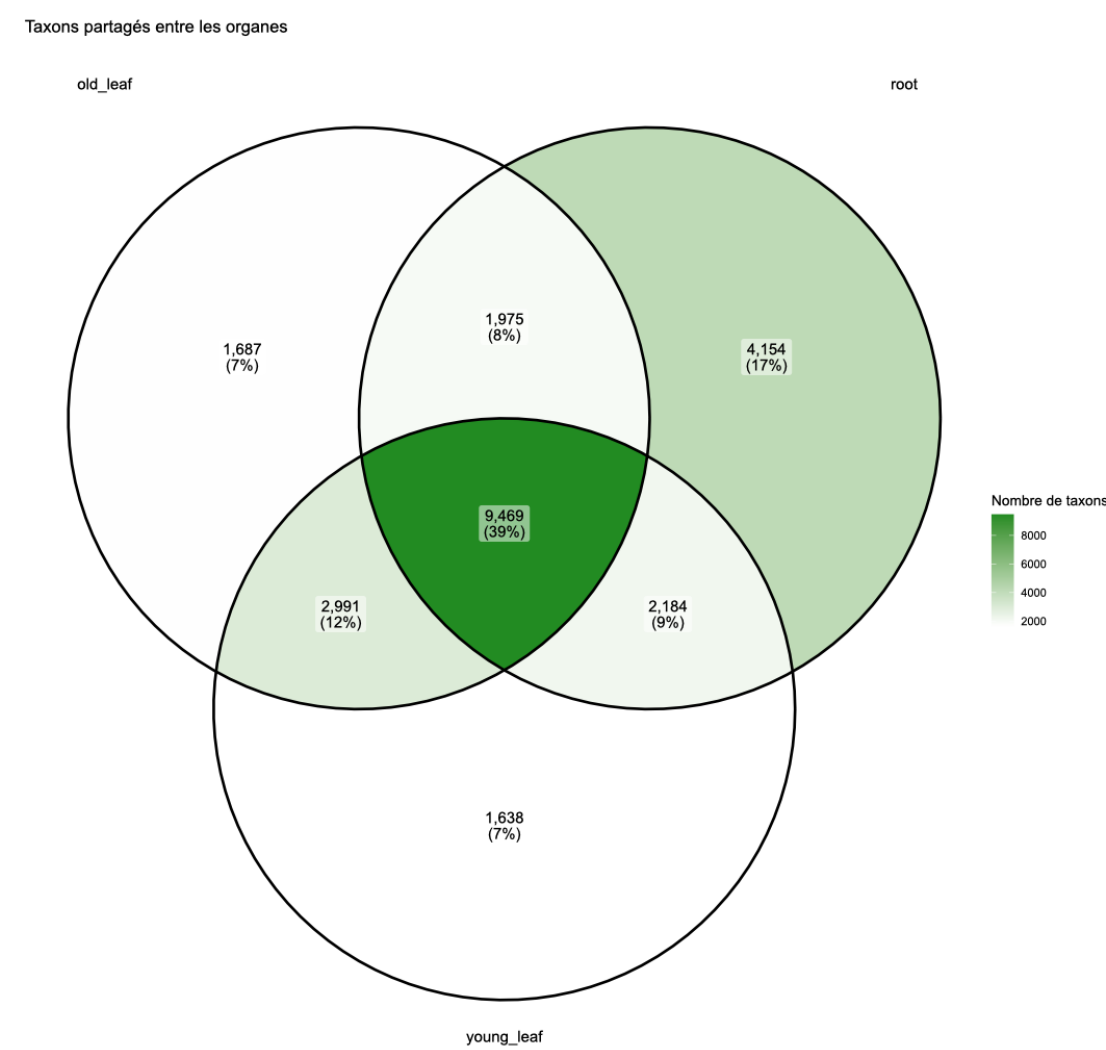
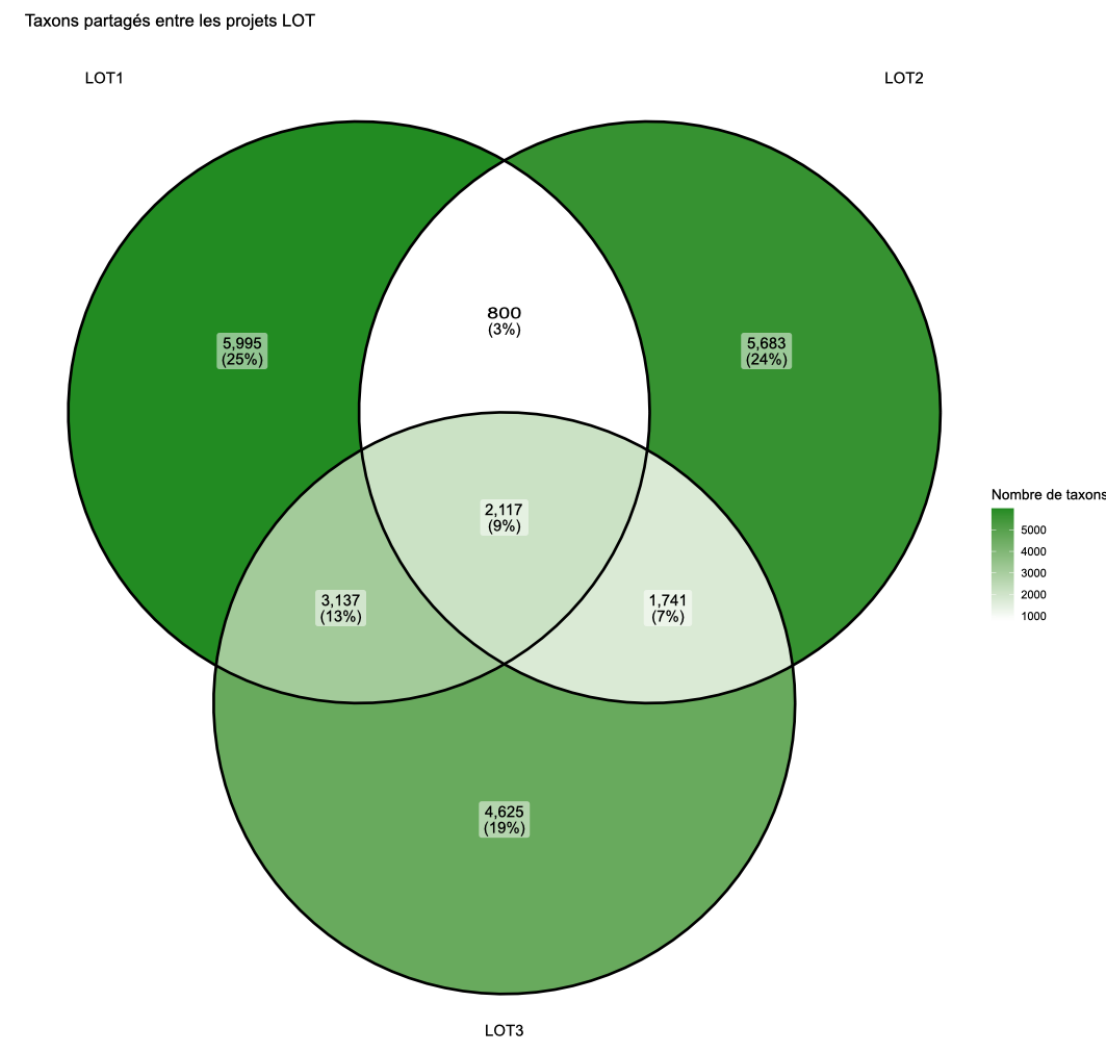
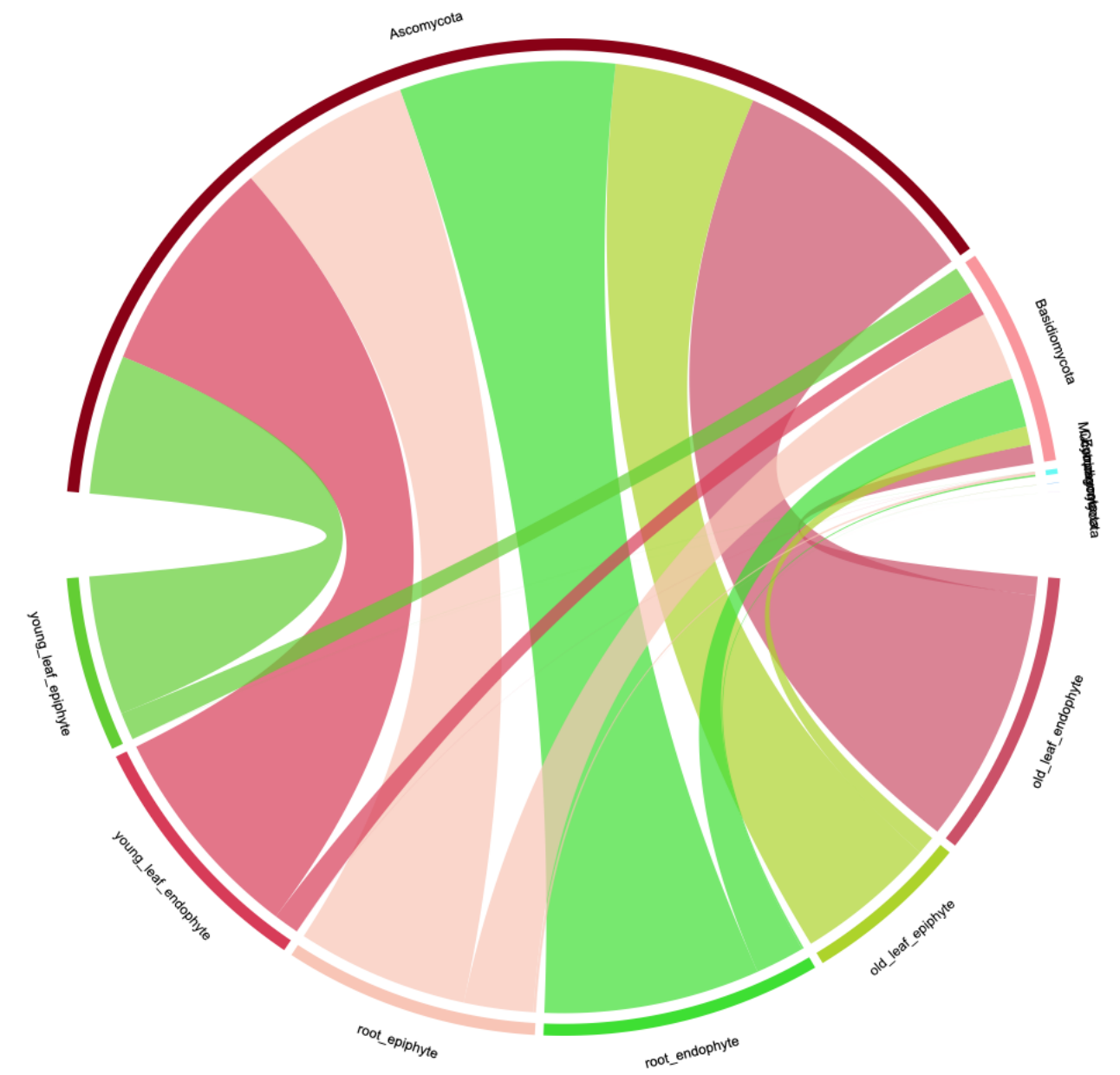
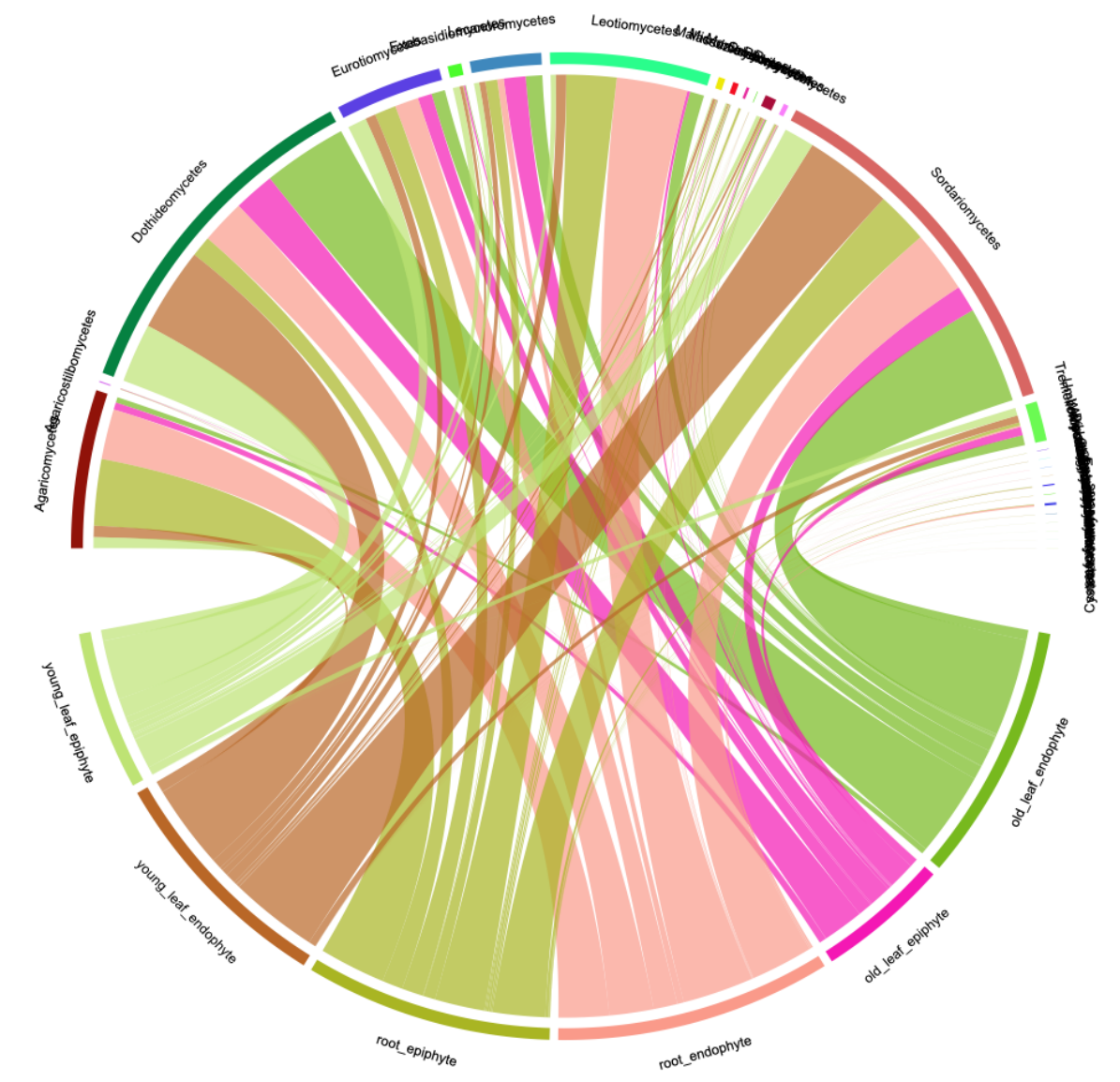
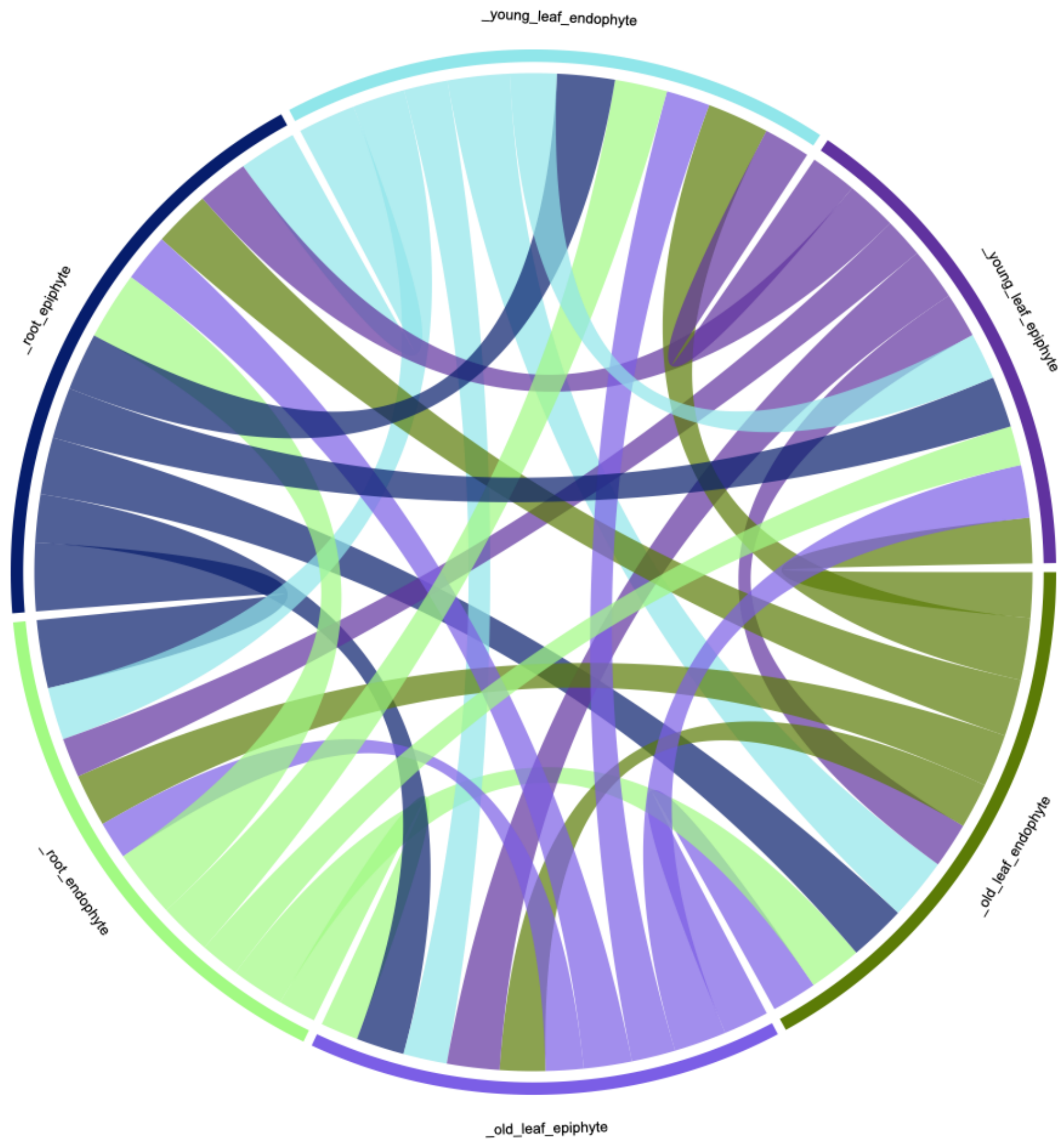


Diagramme de Venn et UpSetR



Diagrammes de Chord



Analyses de diversité Beta

Ordinations (NMDS, k=3, stress=0.13)

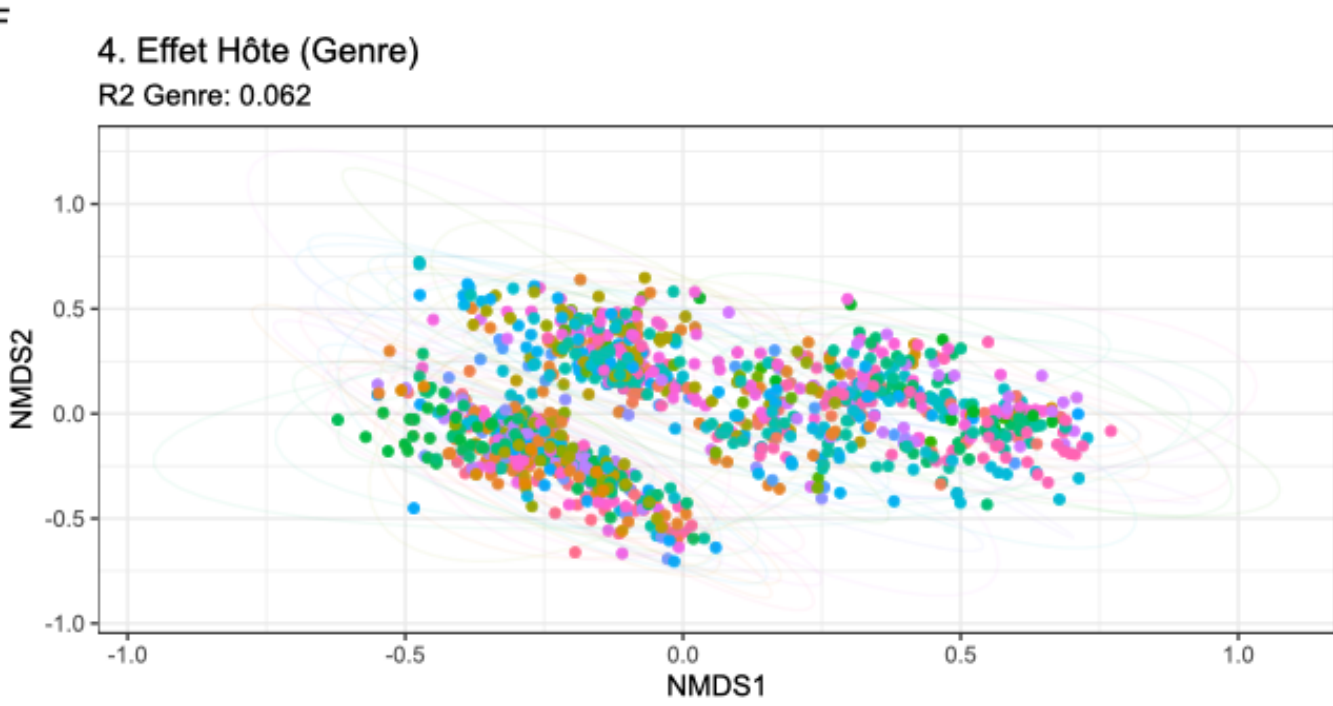
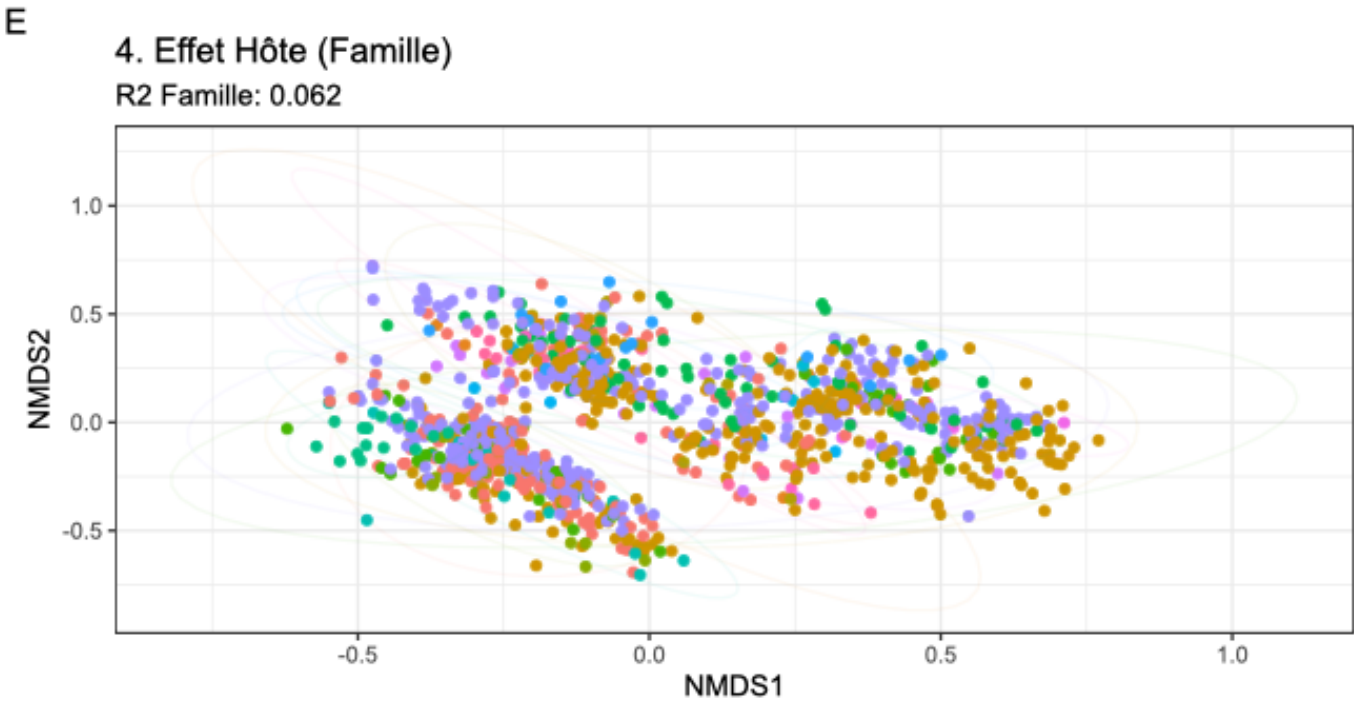
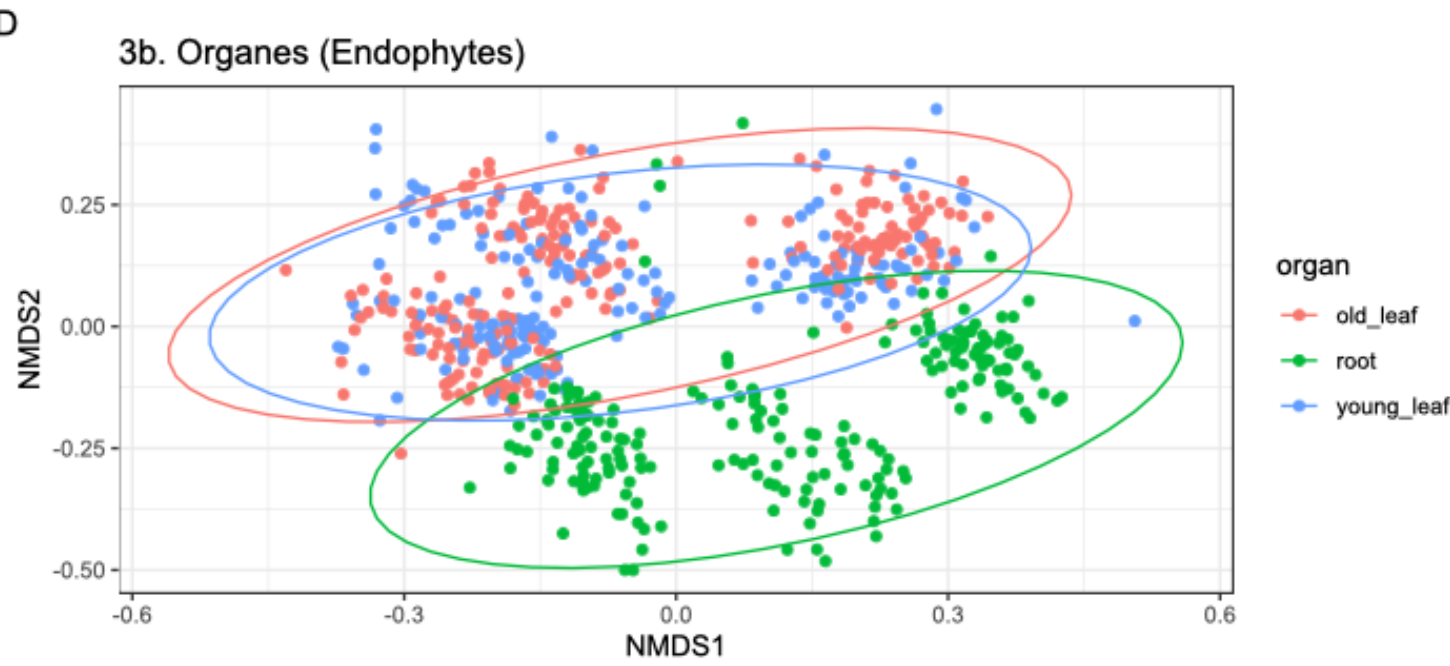
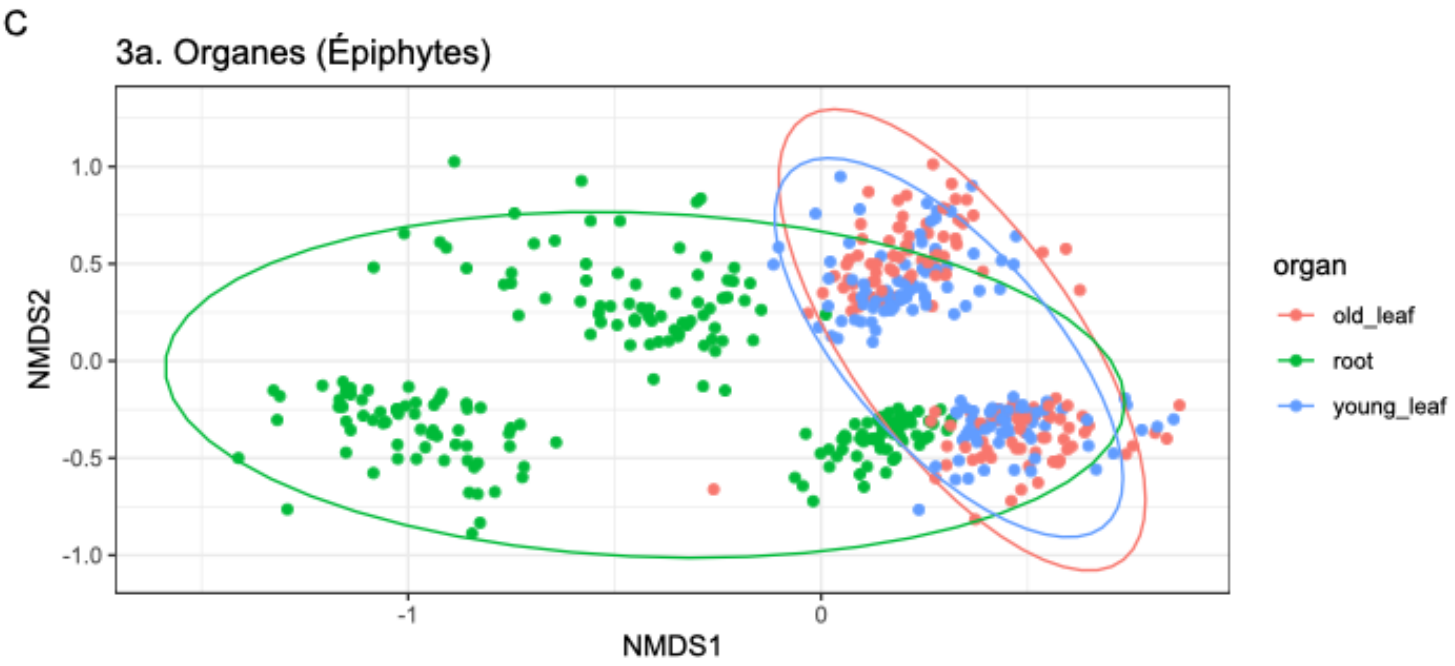
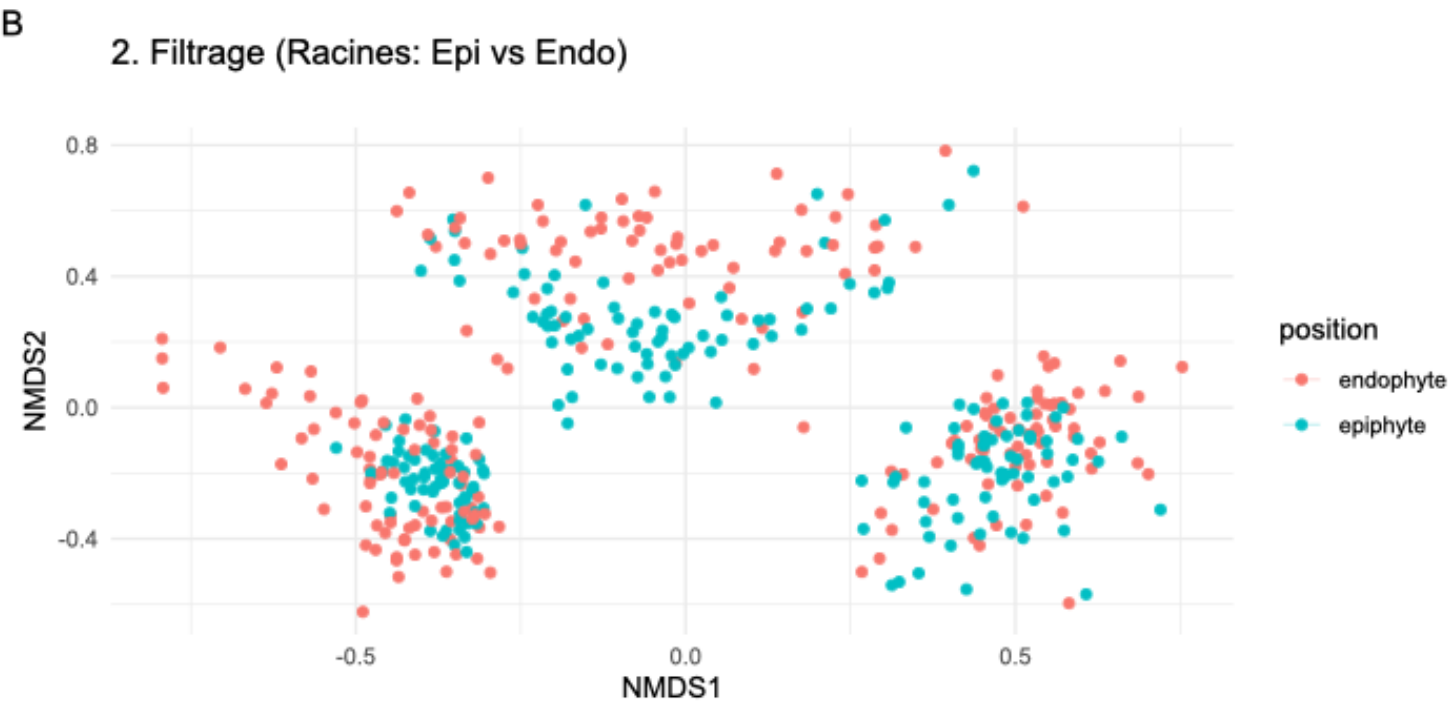
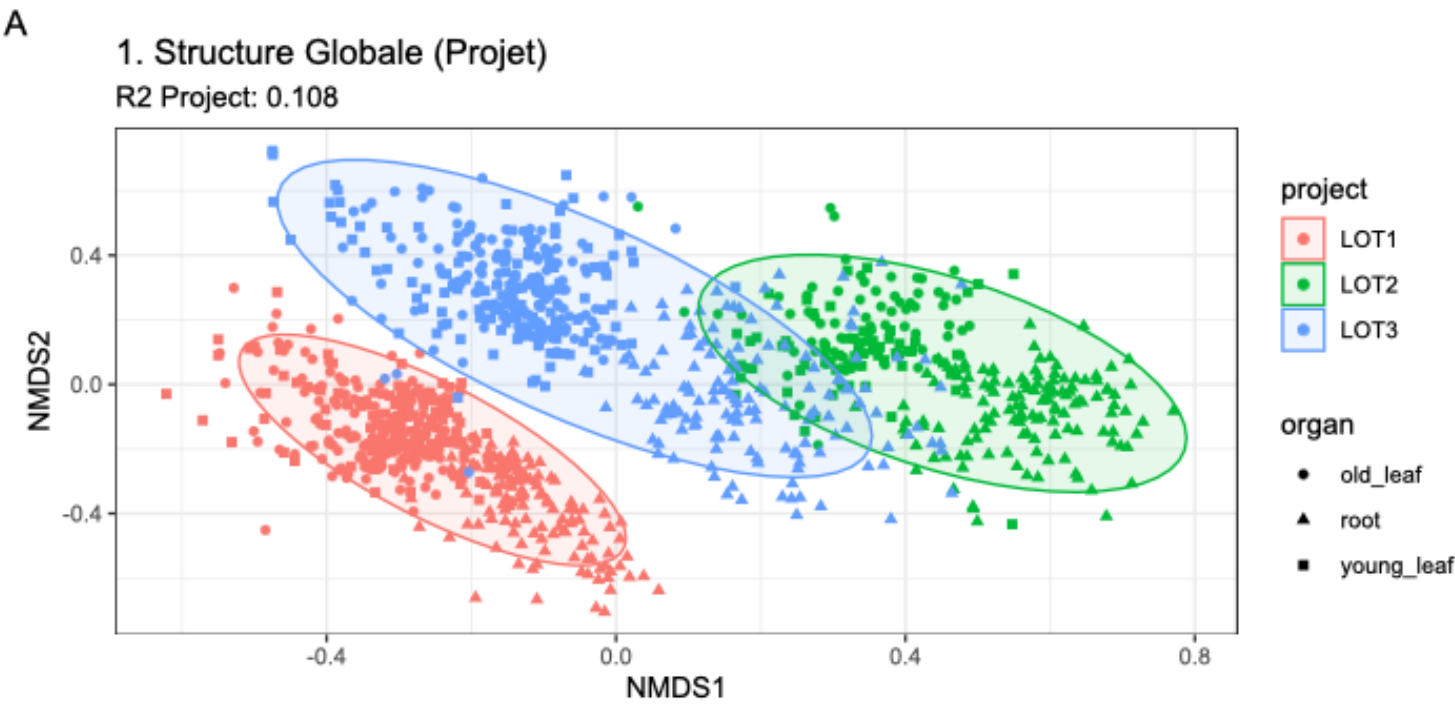
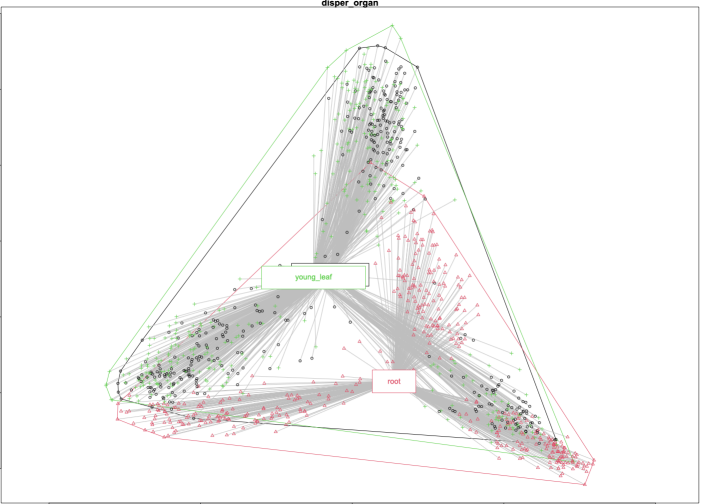
Stress:

- Global : 0.1363989, k=3
- B: 0.1783157, k=2
- C: 0.1592424, k=2
- D: 0.1592424, k=2
- E: 0.2286184, k=2
- F: 0.2286184, k=2

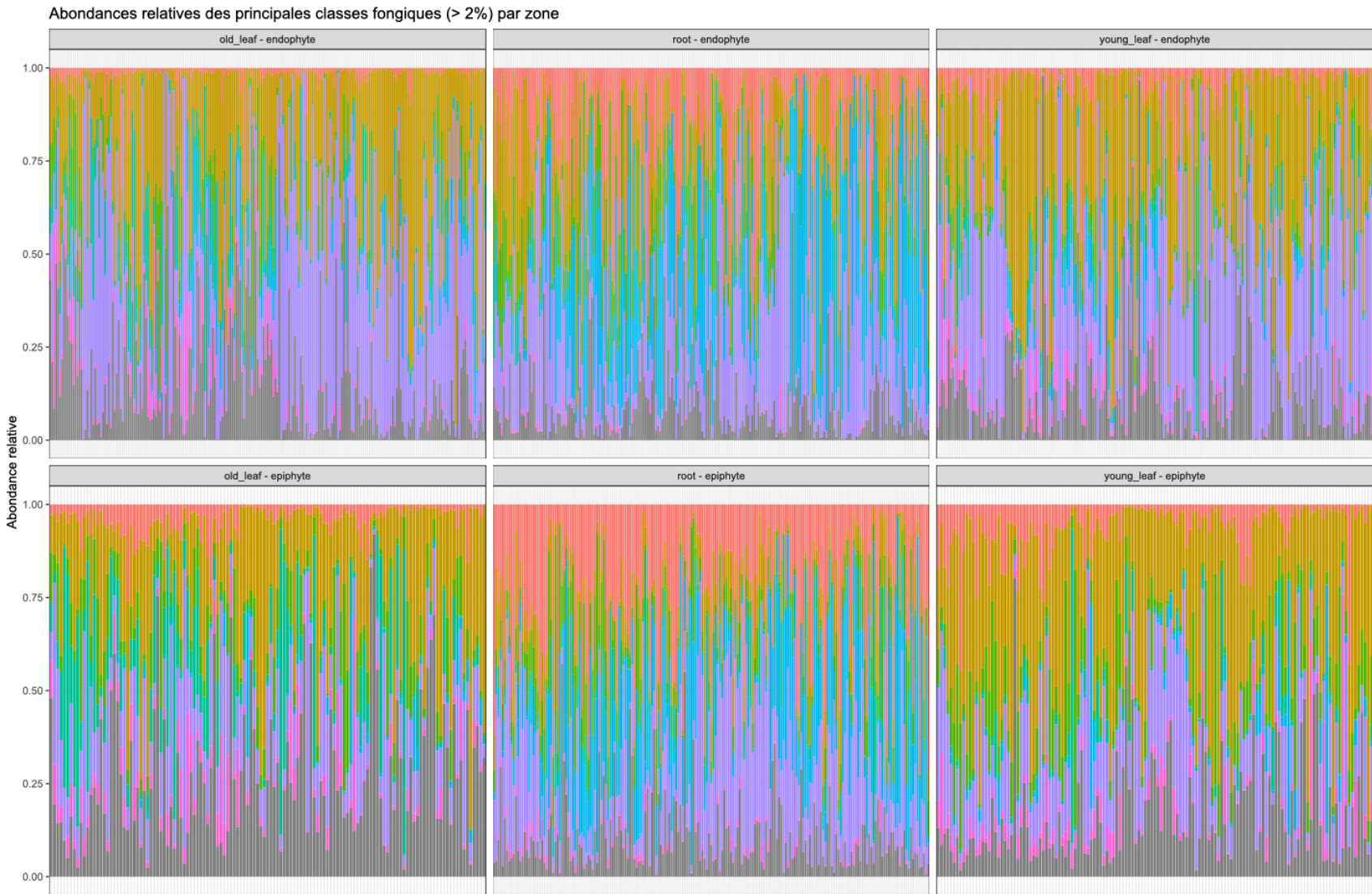
Permanova

	Facteur	Df	R2	p_value
1	project	2	0.1075	0.01
2	organ	2	0.0384	0.01
3	position	1	0.0134	0.01
4	plant_family	16	0.06239	0.01

Dispersions
vérifiées



Abondances relatives des classes fongiques



Classe fongique

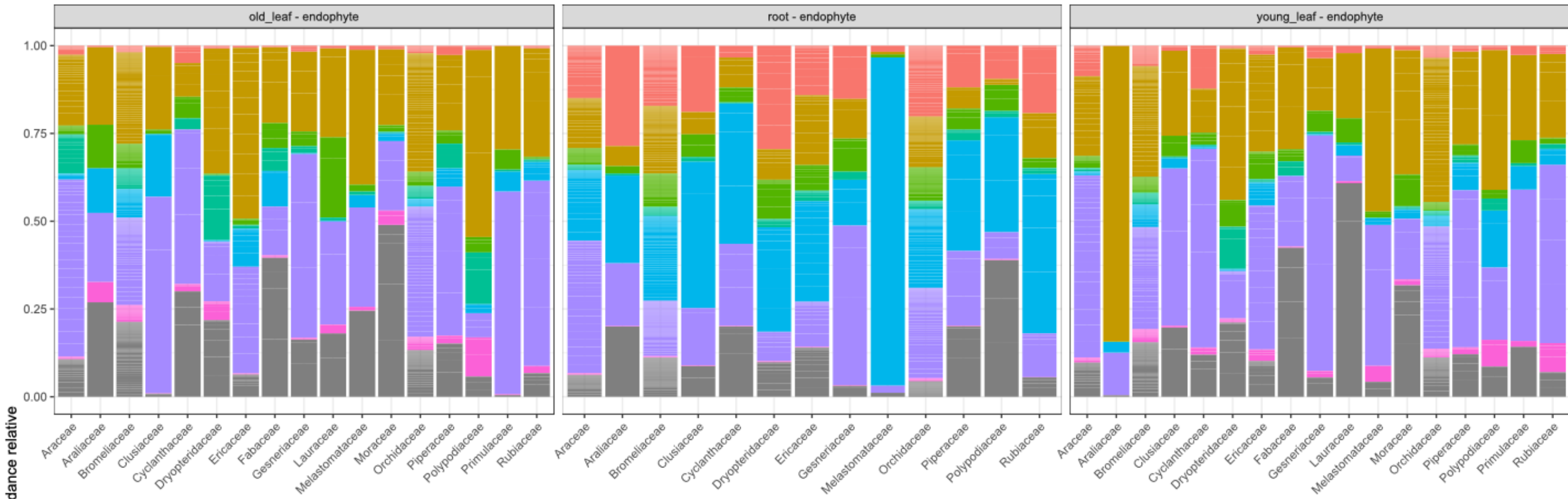
- Agaricomycetes
- Eurotiomycetes
- Leotiomycetes
- Tremellomycetes
- Dothideomycetes
- Lecanoromycetes
- Sordariomycetes
- NA



Classe fongique

- Agaricomycetes
- Eurotiomycetes
- Leotiomycetes
- Tremellomycetes
- Dothideomycetes
- Lecanoromycetes
- Sordariomycetes
- NA

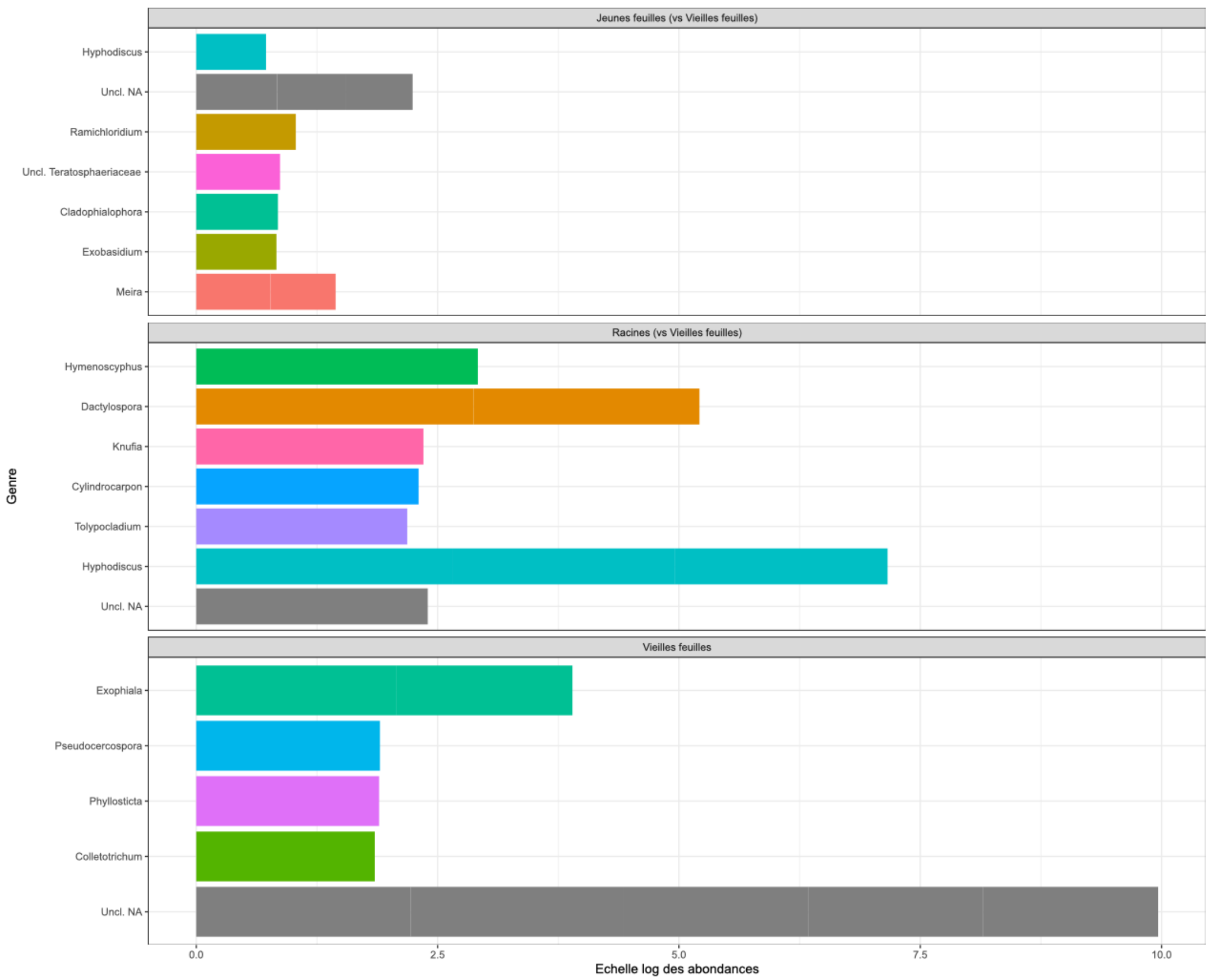
Abondances relatives des principales classes fongiques (> 2%) par famille de plante



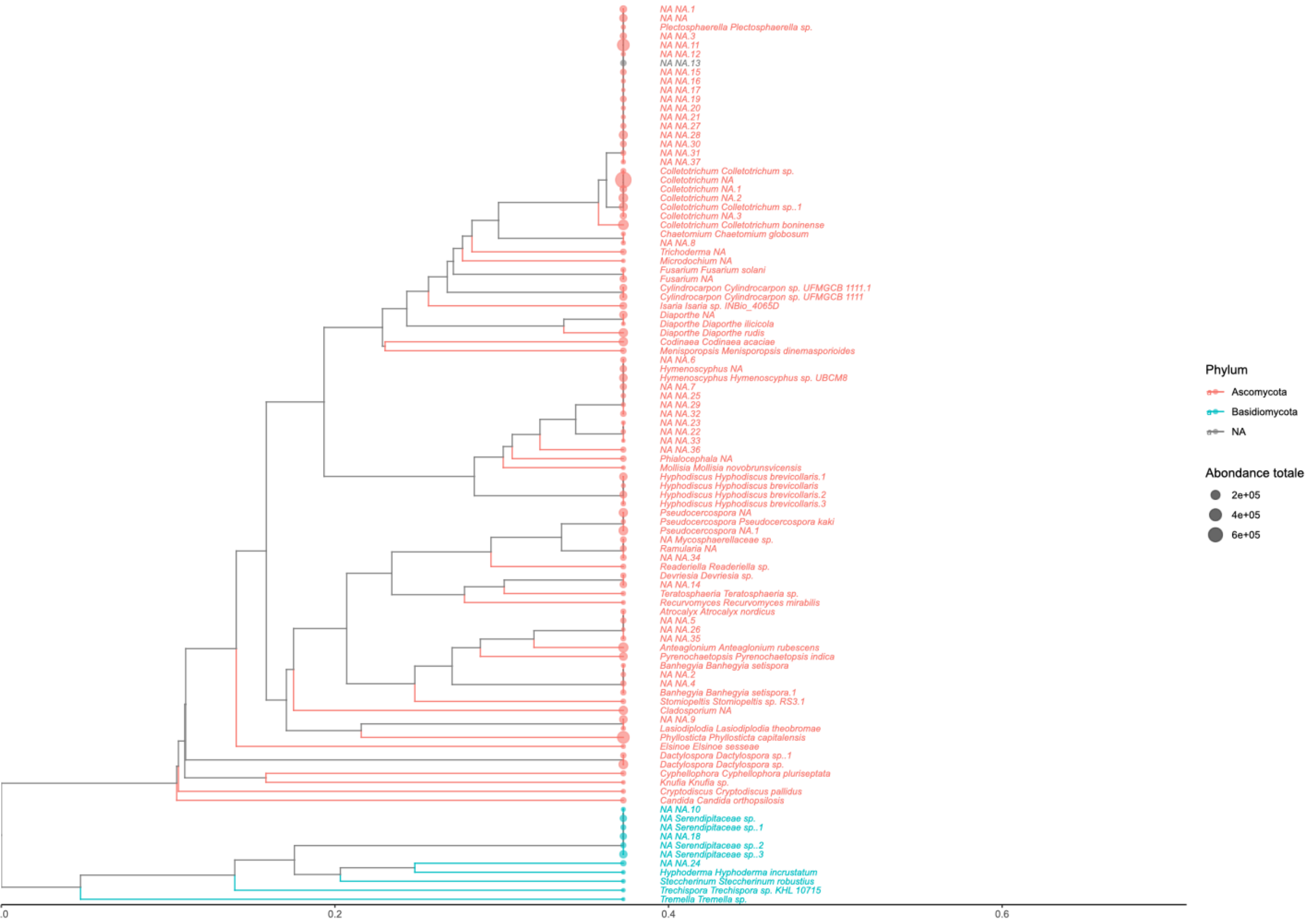
Classe fongique

- Agaricomycetes
- Eurotiomycetes
- Leotiomycetes
- Tremellomycetes
- Dothideomycetes
- Lecanoromycetes
- Sordariomycetes
- NA

Taxons différentiellements abondants entre les organes et arbre des 100 OTU les plus abondants



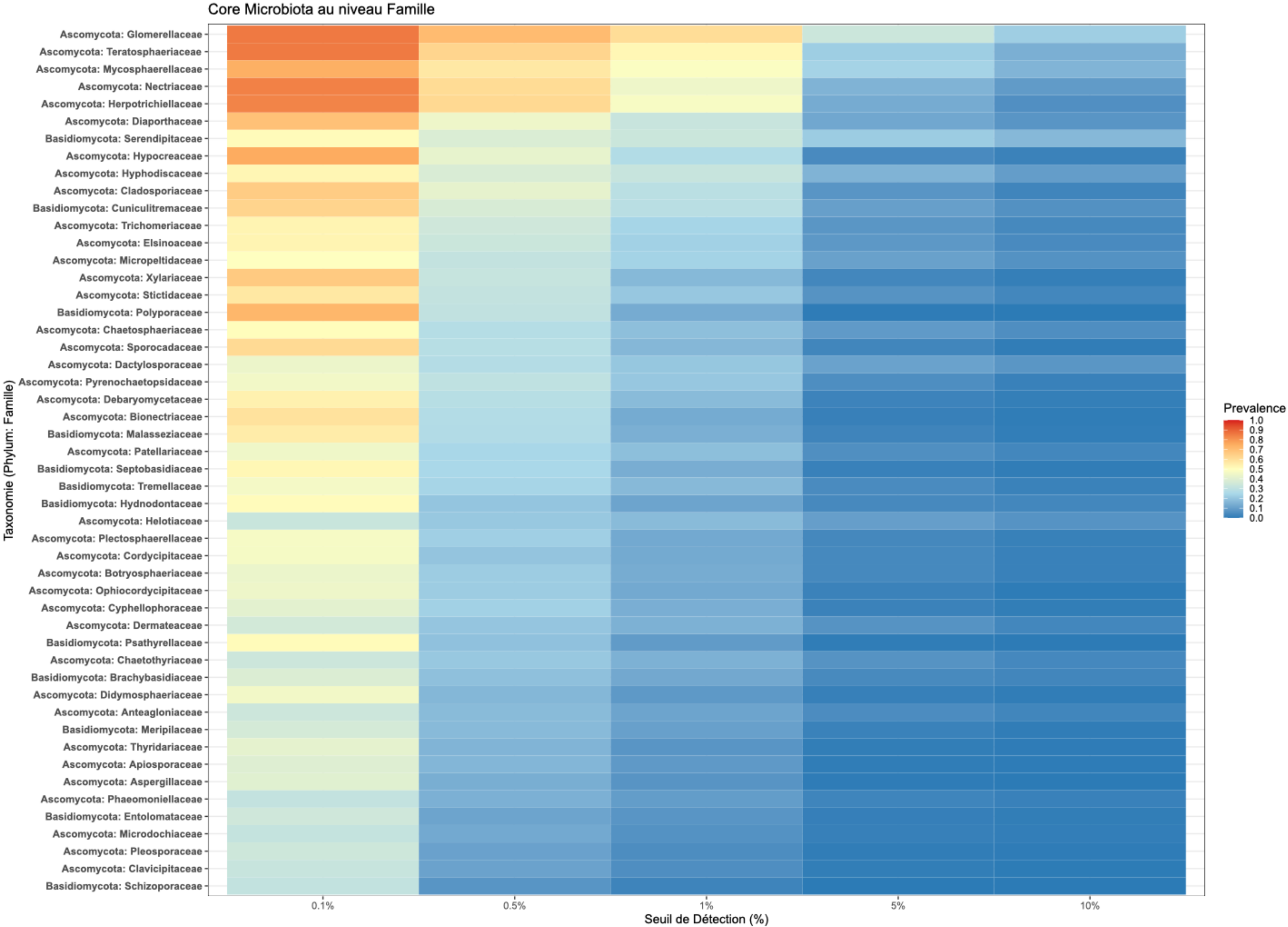
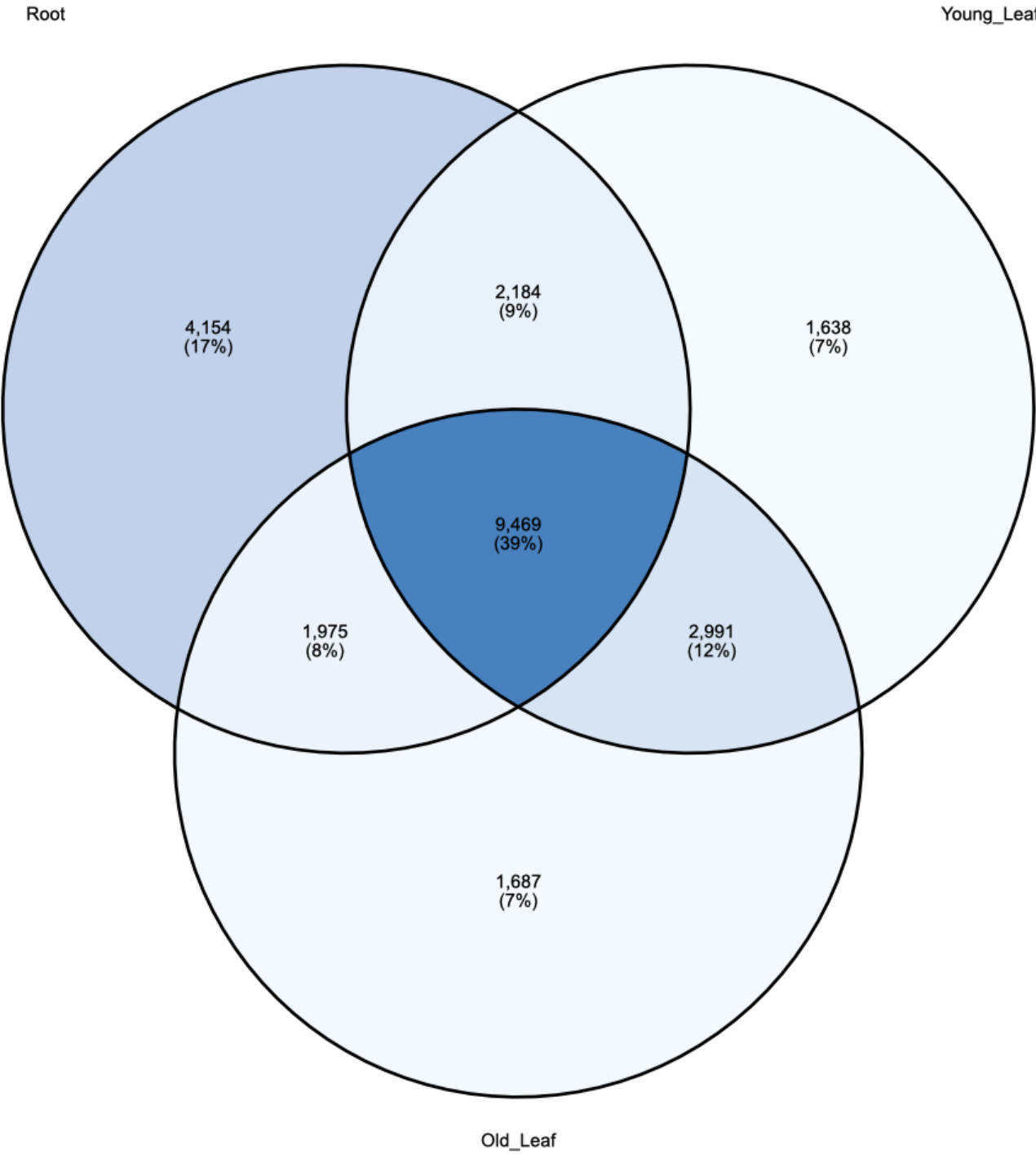
Top 100 OTU



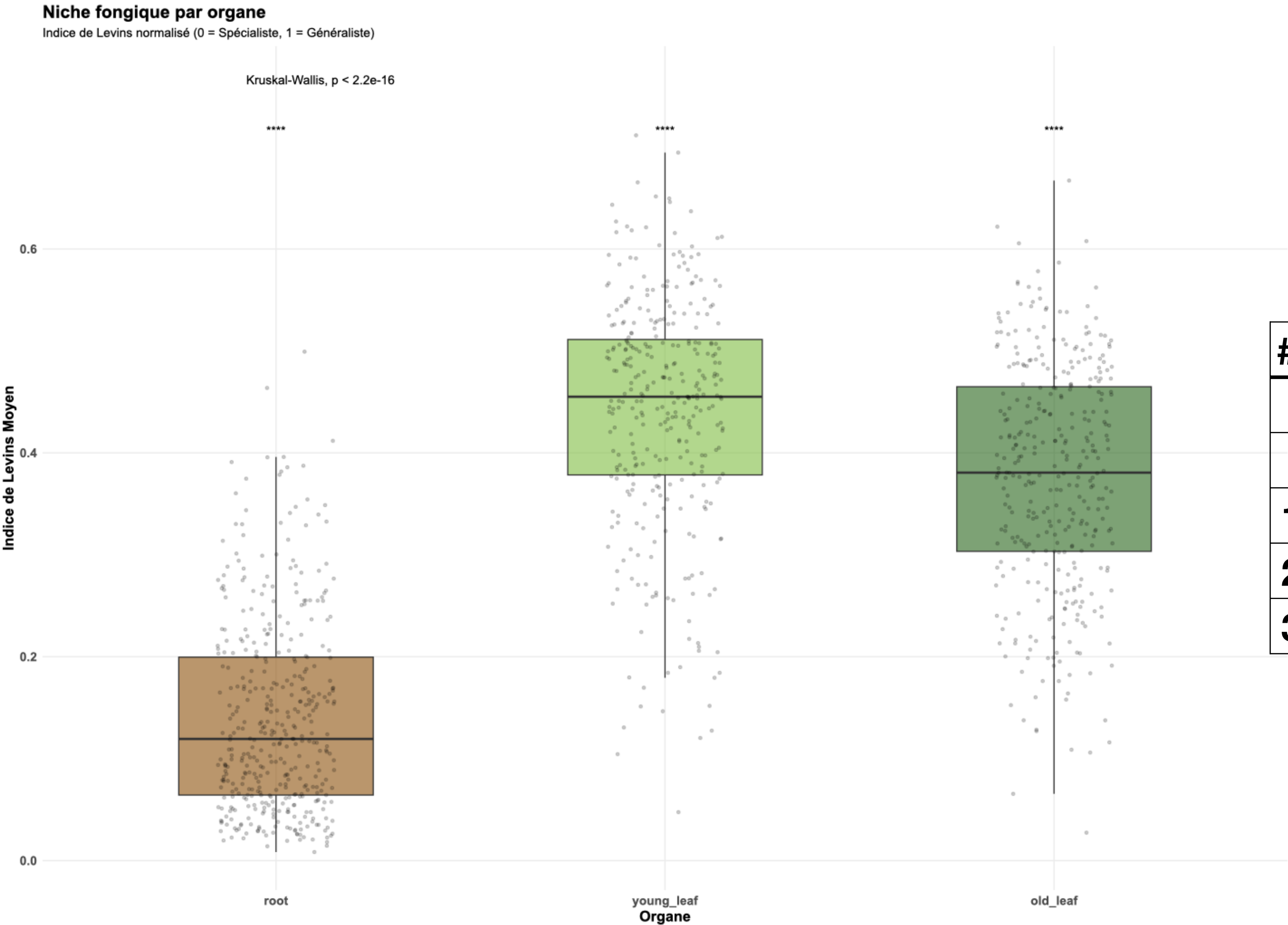
Classé uniquement avec taxonomie, séquences dispo ?

Core microbiota

Partage des OTUs entre les organes



Indice de Levins (B)

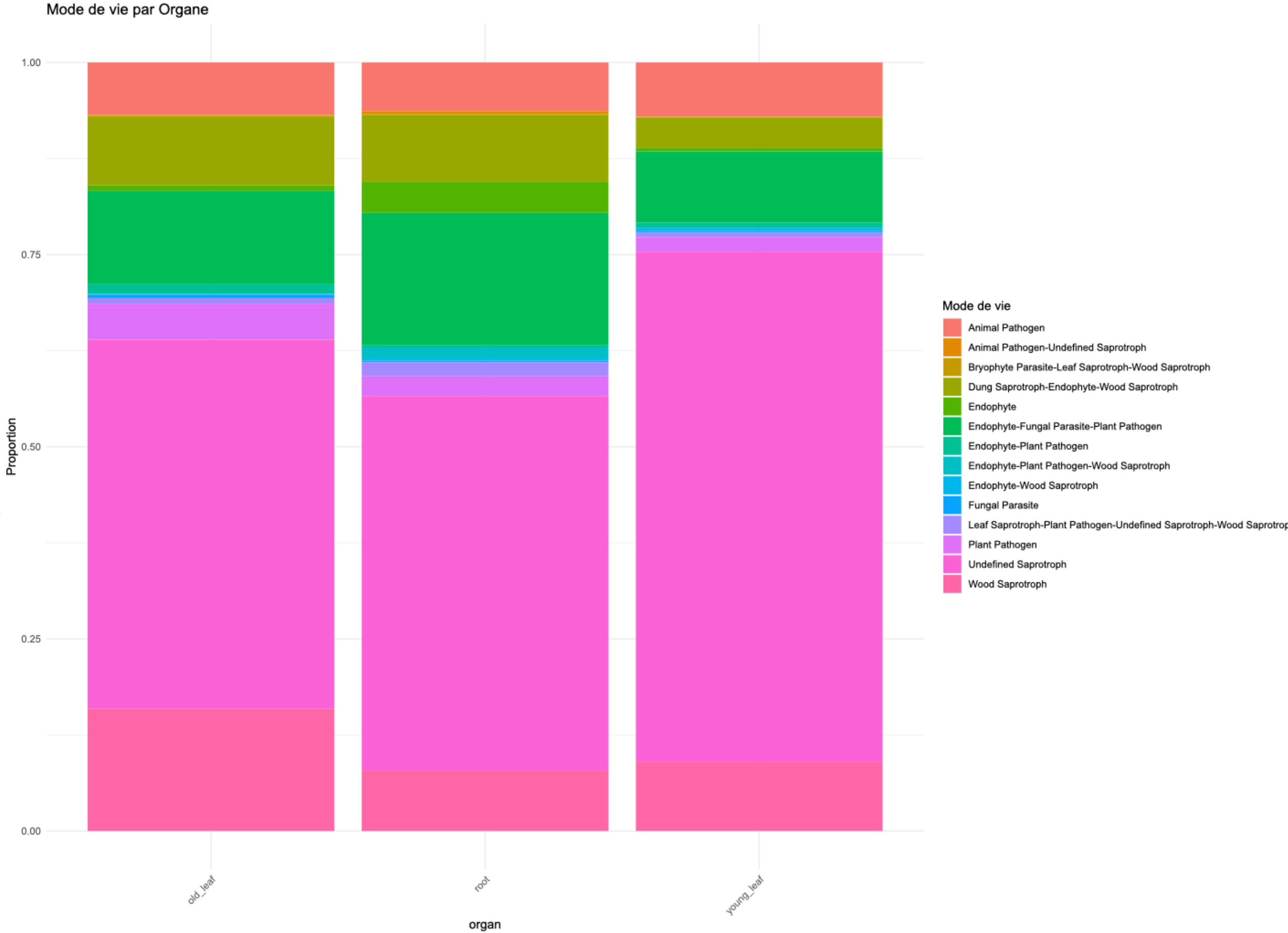
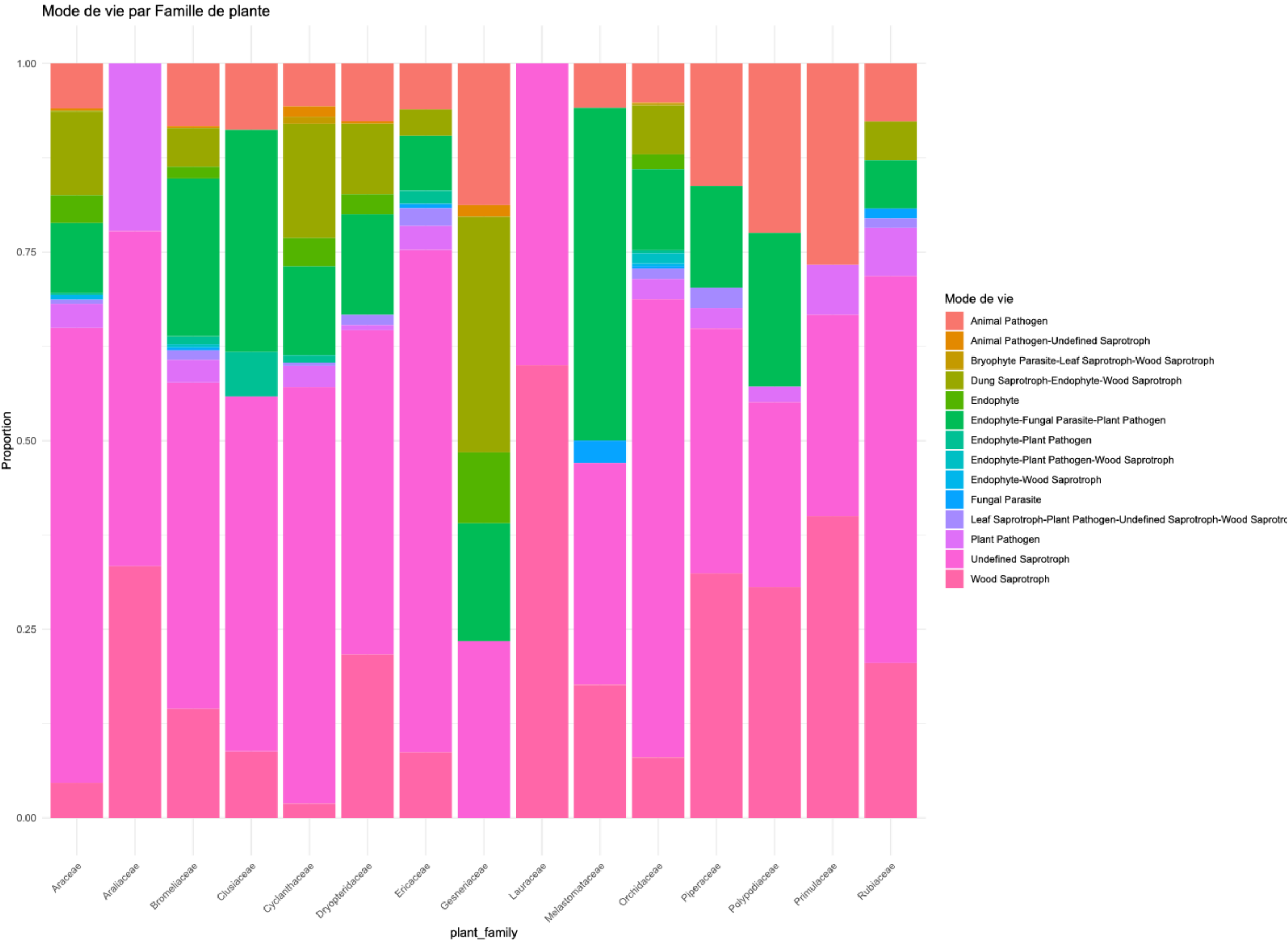


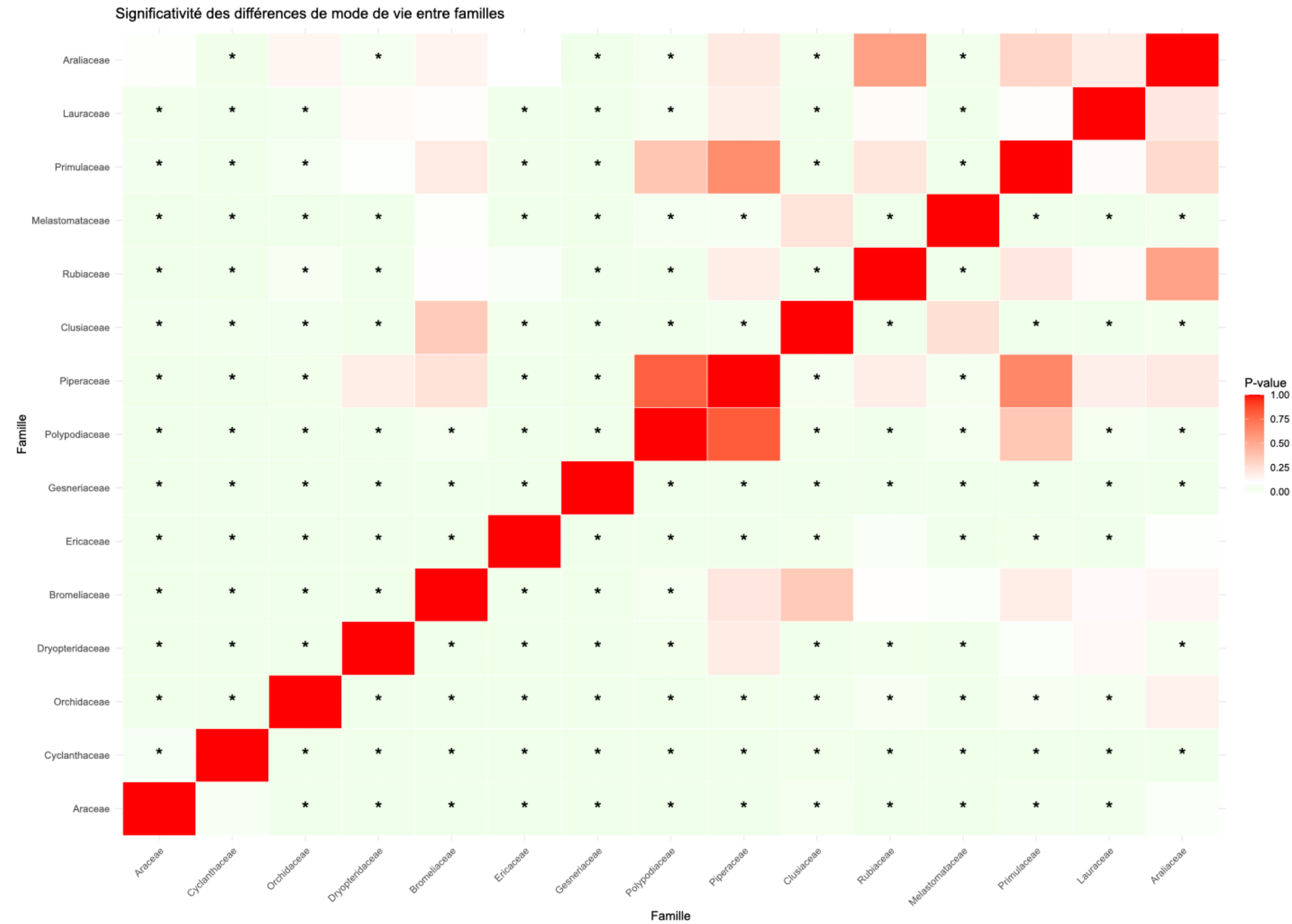
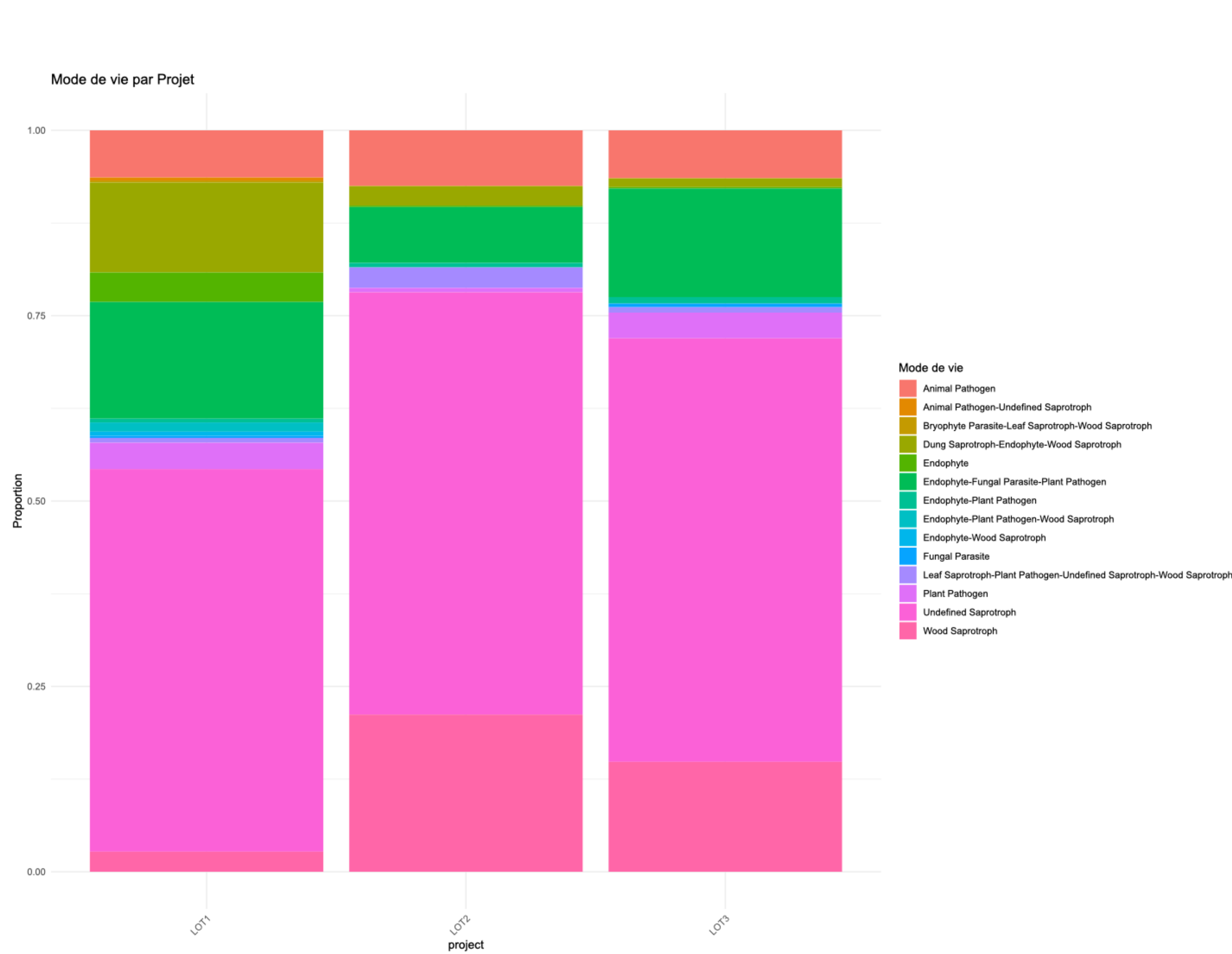
Summary

#	A tibble:	3	× 4		
	organ		Moyenne	SD	n
	<fct>		<dbl>	<dbl>	<int>
1	old_leaf		0.377	0.112	354
2	young_leaf		0.440	0.115	336
3	root		0.140	0.0946	417

Analyse des traits fonctionnels

Utilisation de Fungal Traits





--- Test : Famille vs Mode de vie ---

Pearson's Chi-squared test with simulated p-value (based on 2000 replicates)

data: tbl

X-squared = 1043.7, df = NA, p-value = 0.0004998

Analyse canonique des corrélations (ACC)

```
r$> cca_mod <- cca(df_fonctions_mat ~ organ + project + plant_family, data = metadata_sub)

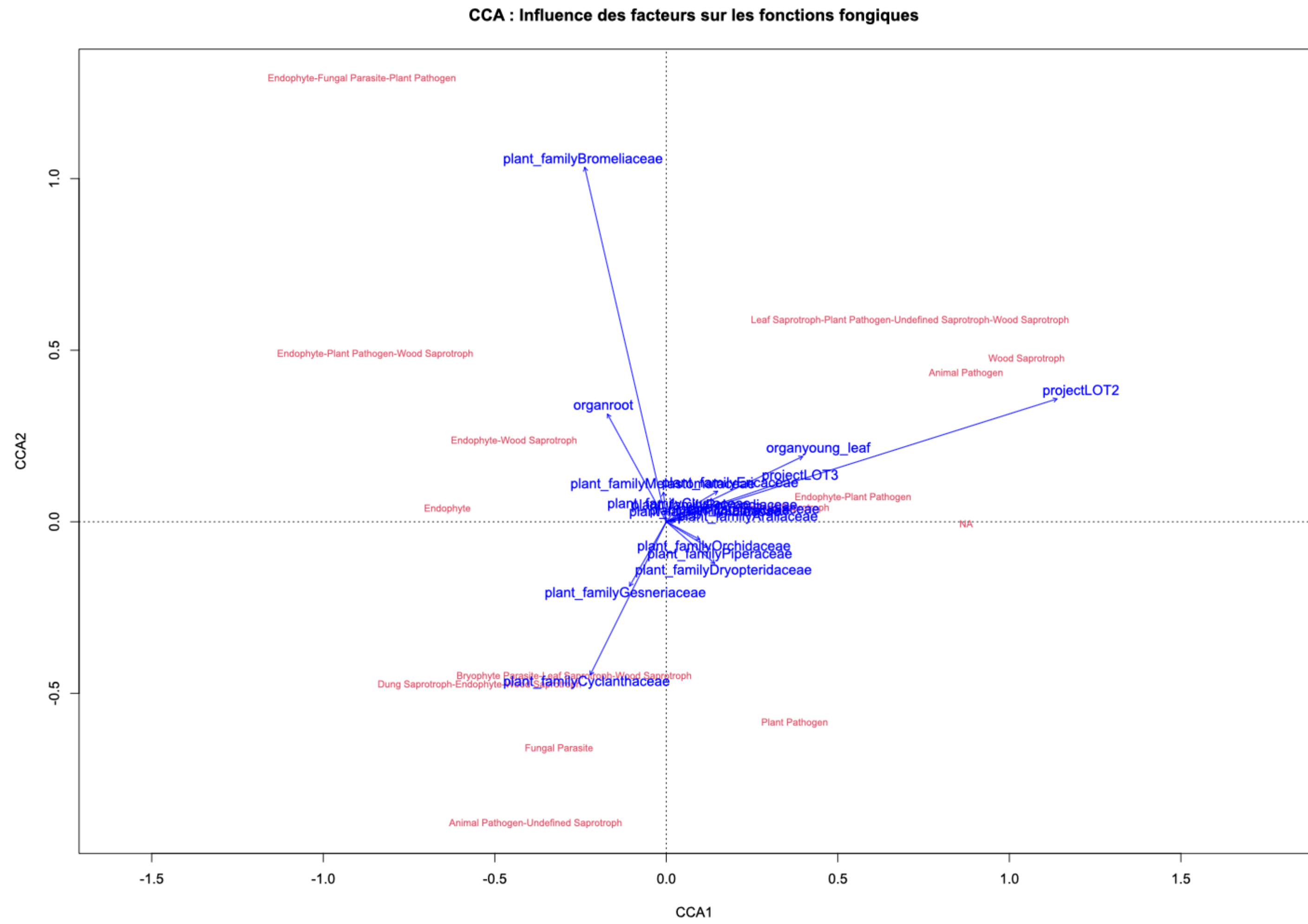
r$> anova(cca_mod, by = "term")
Permutation test for cca under reduced model
Terms added sequentially (first to last)
Permutation: free
Number of permutations: 999

Model: cca(formula = df_fonctions_mat ~ organ + project + plant_family, data = metadata_sub)
      Df ChiSquare      F Pr(>F)
organ    2    0.2813 23.1596 0.001 ***
project  2    0.4907 40.4026 0.001 ***
plant_family 14    0.4861  5.7186 0.220
Residual 962    5.8415

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

r$> print(scores_cca)
```

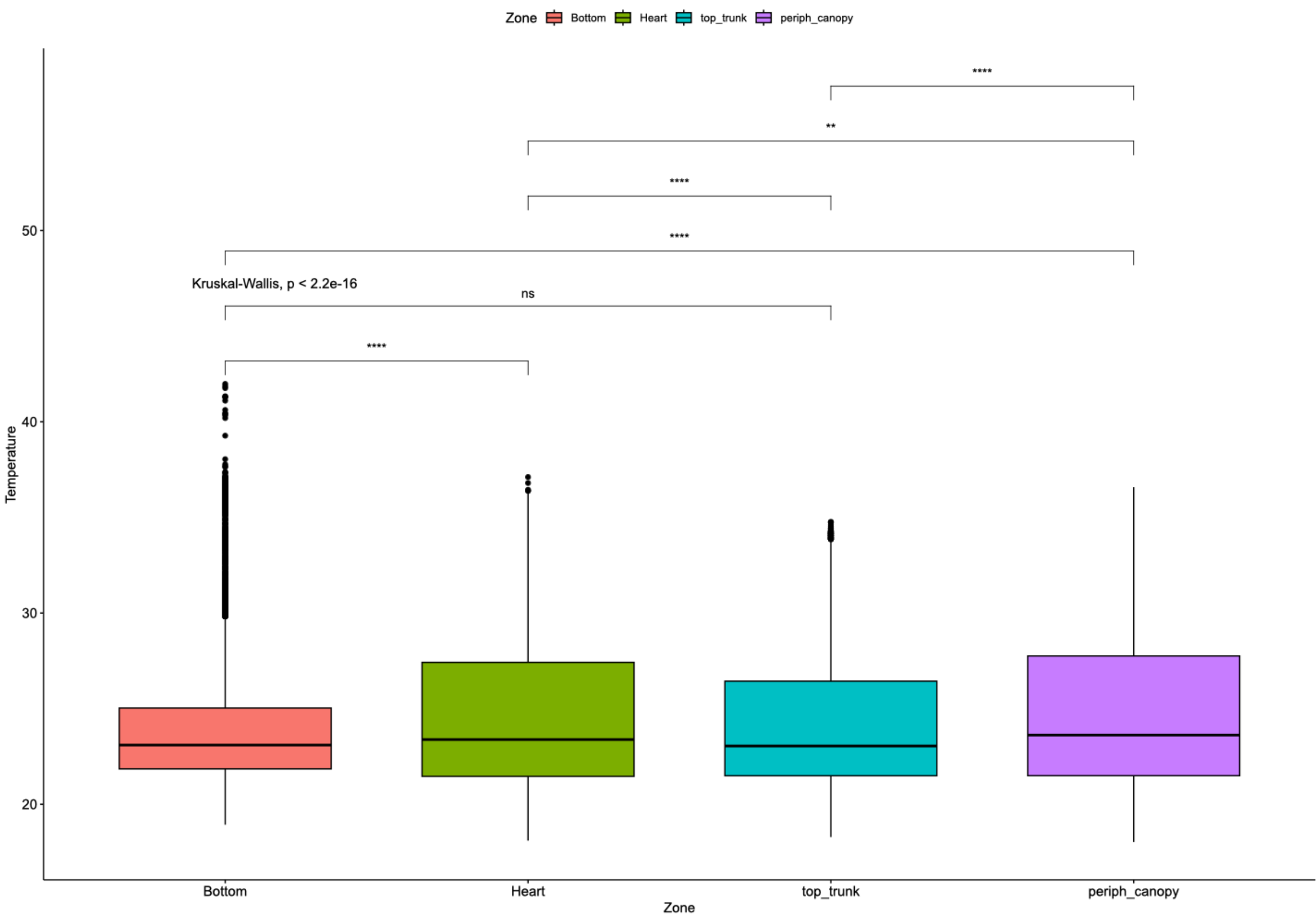
	CCA1	CCA2
organroot	-0.12514726	0.2287148313
organyoung_leaf	0.28950032	0.1386545380
projectLOT2	0.82996800	0.2610134551
projectLOT3	0.23738973	0.0809383144
plant_familyAraliaceae	0.01846631	0.0009955989
plant_familyBromeliaceae	-0.17332678	0.7528486165
plant_familyClusiaceae	0.01520892	0.0207114985
plant_familyCyclanthaceae	-0.16151660	-0.3250022910
plant_familyDryopteridaceae	0.10275298	-0.0885220735
plant_familyEricaceae	0.10897247	0.0649908762
plant_familyGesneriaceae	-0.07796521	-0.1365652442
plant_familyLauraceae	0.03249132	0.0063906642
plant_familyMelastomataceae	-0.00611200	0.0626997446
plant_familyOrchidaceae	0.07098788	-0.0376581651
plant_familyPiperaceae	0.08773575	-0.0542693011
plant_familyPolypodiaceae	0.05382934	0.0178163151
plant_familyPrimulaceae	0.01515549	0.0035934823
plant_familyRubiaceae	0.06352557	0.0090436077



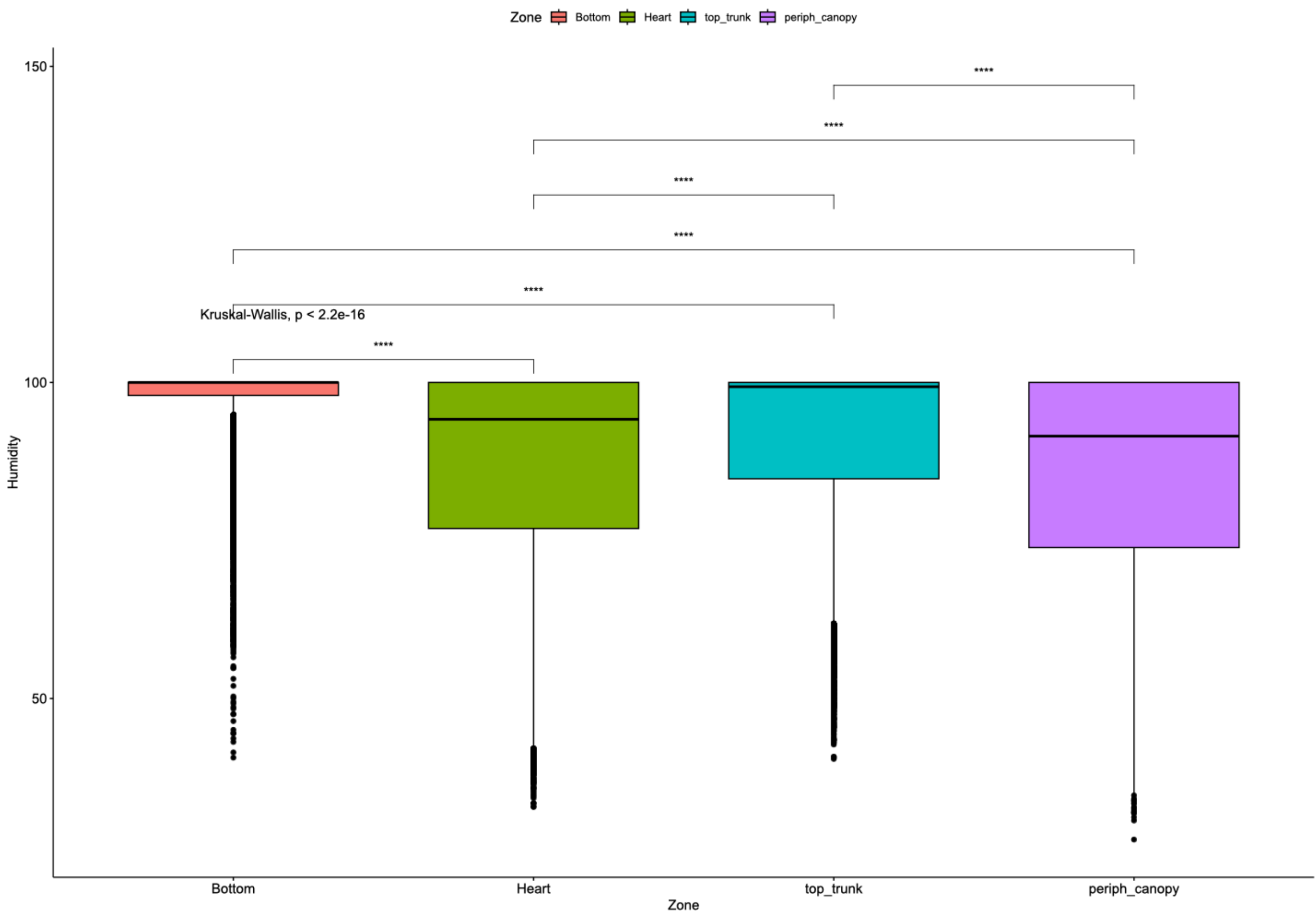
Croisement phyloseq x data env

Avec les données du LOT 1 - diff. env. / zones

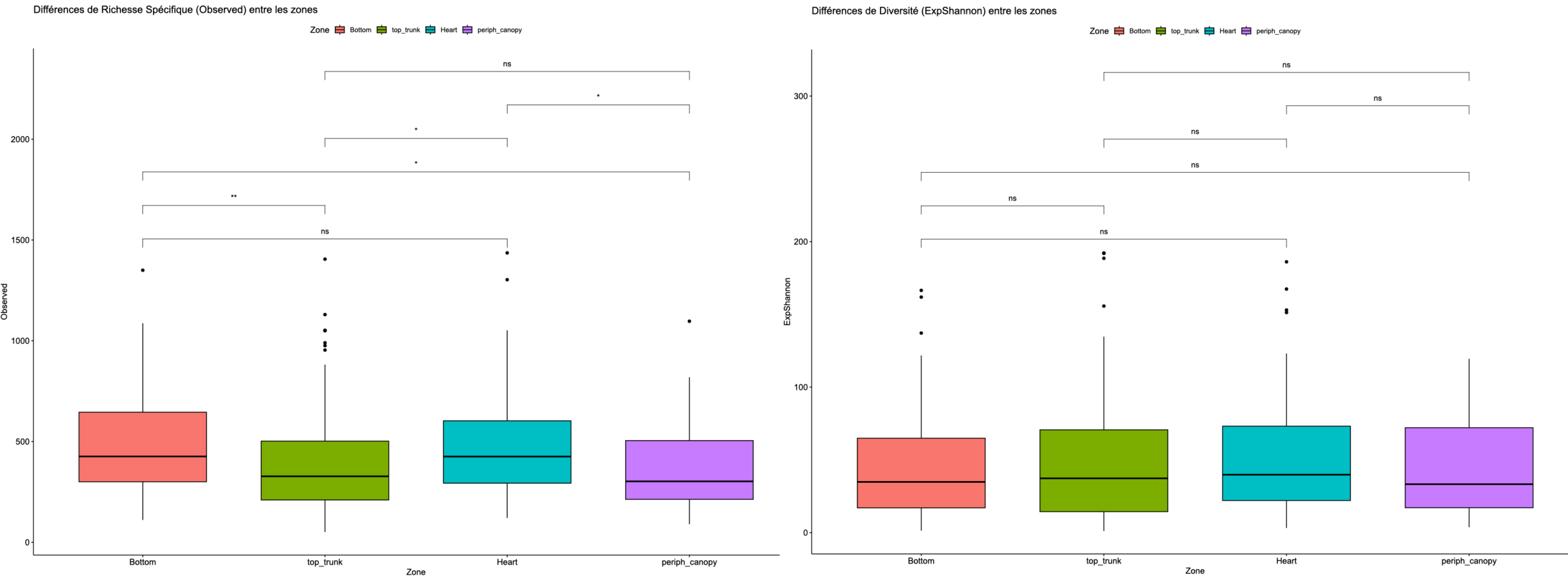
Différences de Température entre les zones



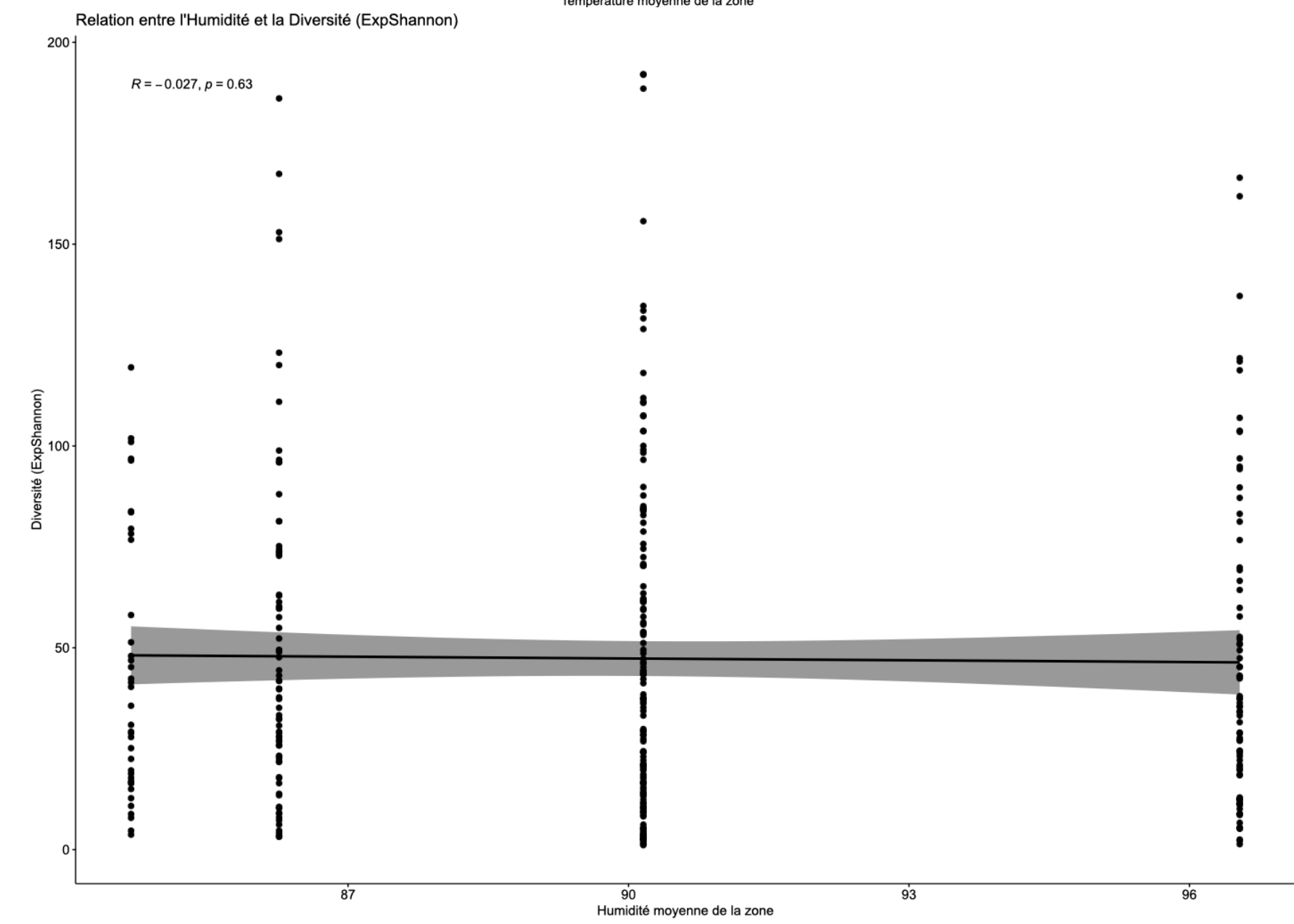
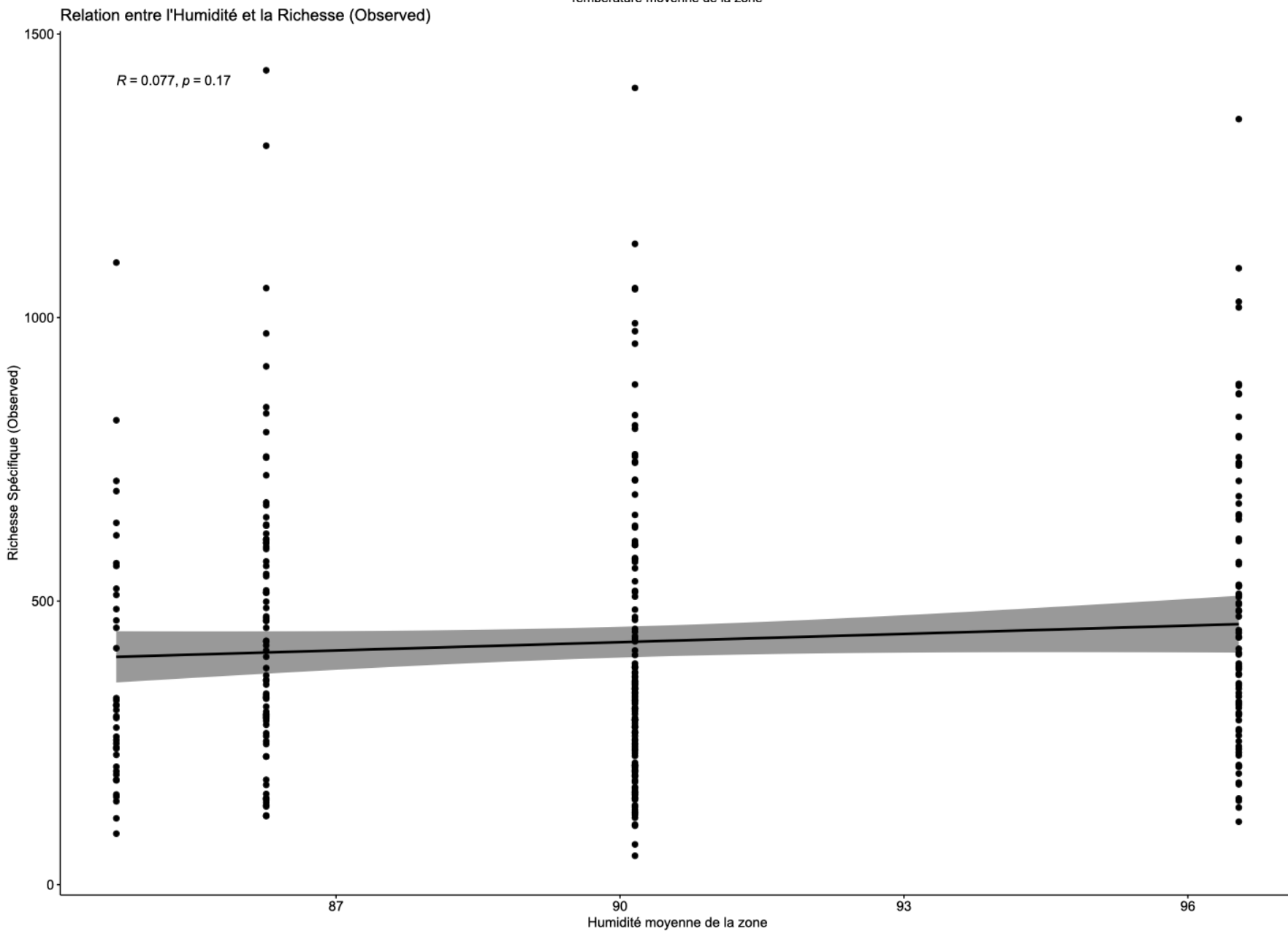
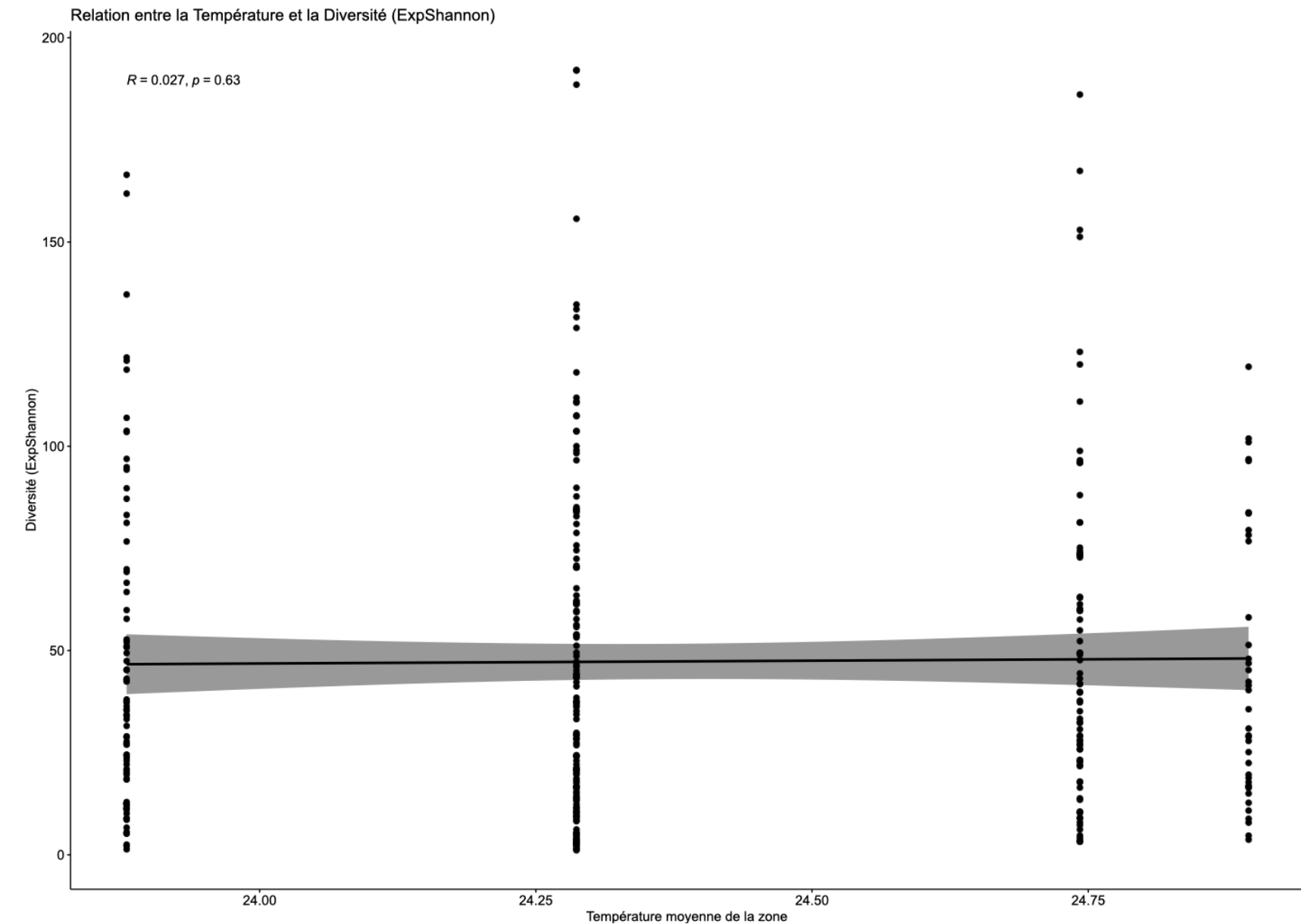
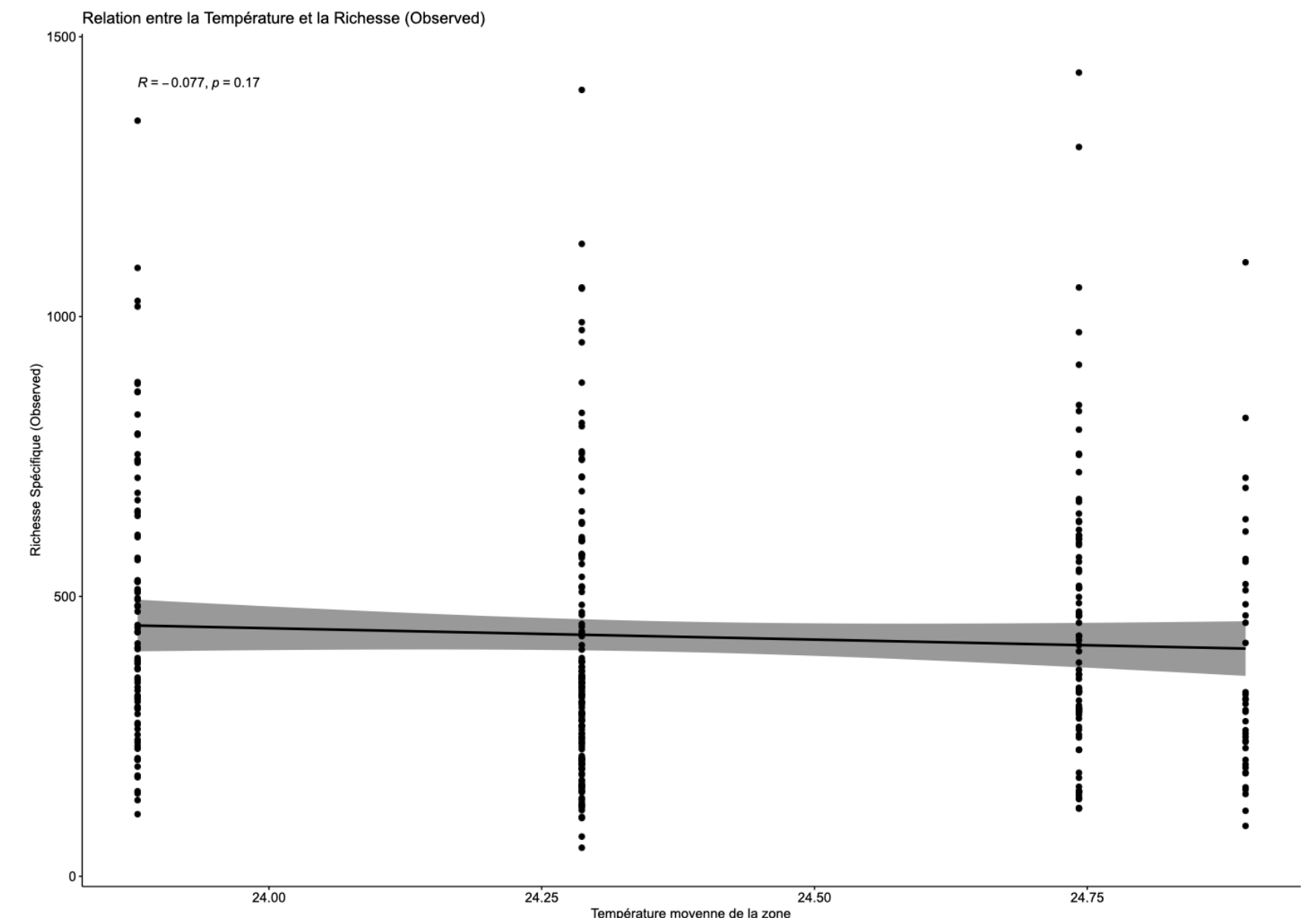
Différences d'Humidité entre les zones



Differences de q=0 et q=1 par zones



Correlations q avec env



Test avec glm (négative binomial)

```
r$> glm_humi_temp_obs <- glm.nb(Observed ~ Mean_Humi * Mean_Temp, data = df_richness)
r$> summary(glm_humi_temp_obs)

Call:
glm.nb(formula = Observed ~ Mean_Humi * Mean_Temp, data = df_richness,
       init.theta = 3.323696435, link = log)

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)    262.70346   139.57495    1.882   0.0598 .
Mean_Humi       -4.03583    1.97777   -2.041   0.0413 *
Mean_Temp      -12.01146    6.24851   -1.922   0.0546 .
Mean_Humi:Mean_Temp  0.18215    0.08715    2.090   0.0366 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for Negative Binomial(3.3237) family taken to be 1)

Null deviance: 357.58  on 329  degrees of freedom
Residual deviance: 346.14  on 326  degrees of freedom
AIC: 4477.3

Number of Fisher Scoring iterations: 1

              Theta:  3.324
            Std. Err.:  0.249

2 x log-likelihood:  -4467.302
```

```
r$> simulationOutput_humi_temp_obs <- simulateResiduals(fittedModel = glm_humi_temp_obs, plot = TRUE)
testUniformity(simulationOutput_humi_temp_obs)
testDispersion(simulationOutput_humi_temp_obs)
testZeroInflation(simulationOutput_humi_temp_obs)

Asymptotic one-sample Kolmogorov-Smirnov test

data:  simulationOutput$scaledResiduals
D = 0.054909, p-value = 0.2728
alternative hypothesis: two-sided

DHARMA nonparametric dispersion test via sd of residuals fitted vs. simulated

data:  simulationOutput
dispersion = 1.0844, p-value = 0.352
alternative hypothesis: two.sided

DHARMA zero-inflation test via comparison to expected zeros with simulation under H0 = fitted model

data:  simulationOutput
ratioObsSim = NaN, p-value = 1
alternative hypothesis: two.sided
```

```
r$> glm_humi_temp_exp <- glm.nb(ExpShannon ~ Mean_Humi * Mean_Temp, data = df_richness)
summary(glm_humi_temp_exp)
There were 50 or more warnings (use warnings() to see the first 50)

Call:
glm.nb(formula = ExpShannon ~ Mean_Humi * Mean_Temp, data = df_richness,
       init.theta = 1.352430258, link = log)

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)    179.6177   221.1007    0.812   0.417
Mean_Humi       -2.4297    3.1329   -0.776   0.438
Mean_Temp       -7.7928    9.8981   -0.787   0.431
Mean_Humi:Mean_Temp  0.1062    0.1380    0.769   0.442

(Dispersion parameter for Negative Binomial(1.3524) family taken to be 1)

Null deviance: 366.77  on 329  degrees of freedom
Residual deviance: 366.04  on 326  degrees of freedom
AIC: 3206.6

Number of Fisher Scoring iterations: 1

              Theta:  1.352
            Std. Err.:  0.100

2 x log-likelihood:  -3196.553
```

```
r$> simulationOutput_humi_temp_exp <- simulateResiduals(fittedModel = glm_humi_temp_exp, plot = TRUE)
testUniformity(simulationOutput_humi_temp_exp)
testDispersion(simulationOutput_humi_temp_exp)
testZeroInflation(simulationOutput_humi_temp_exp)

Asymptotic one-sample Kolmogorov-Smirnov test

data:  simulationOutput$scaledResiduals
D = 0.033697, p-value = 0.8478
alternative hypothesis: two-sided

DHARMA nonparametric dispersion test via sd of residuals fitted vs. simulated

data:  simulationOutput
dispersion = 0.92064, p-value = 0.64
alternative hypothesis: two.sided

DHARMA zero-inflation test via comparison to expected zeros with simulation under H0 = fitted model

data:  simulationOutput
ratioObsSim = 0, p-value = 0.16
alternative hypothesis: two.sided
```

Graphiques d'interactions

